Swarm Intelligence

CSE-2530: Computational Intelligence

Stephanie Tan

(Slides adapted from Joost Broekens; and credits to Judith Redi and Pradeep K. Murukannaiah, Luciano Cavalcante Siebert)

Last week…

- Evolutionary Algorithms (next week)
  - Inspired by biological evolution
  - Population-based random search from generation to generation

Next

- Swarm intelligence
  - A form of distributed optimization
  - … more “collective” than Genetic Algorithms
- Ant Colony Optimization (ACO)

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Learning Objectives (for this lecture)

By the end of this lecture, you should be able to:

- Understand **swarm intelligence** as a form of distributed optimization
- Explain the **Ant Colony Optimization** (ACO) algorithm
- Apply ACO to the **Traveling Salesman Problem**

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Study Material

- Not in the text book(s)!
- Lecture slides
- Ant Colony Optimization:
  - Reading material:
    - Concept: Bonabeau, E., & Théraulaz, G. (2000). **Swarm smarts**. Scientific American, 282(3), 72-79.
    - Algorithm: Colorni, A., Dorigo, M., & Maniezzo, V. (1991, December). **Distributed optimization by ant colonies**. In Proceedings of the first European conference on artificial life (Vol. 142, pp. 134-142).
  - Optional:
    - Book: Dorigo, M., & Stutzle, T. (2004). **Ant colony optimization.** The MIT Press. 321p. https://tudelft.on.worldcat.org/oclc/57182707 (eBook available through TU Delft student account)

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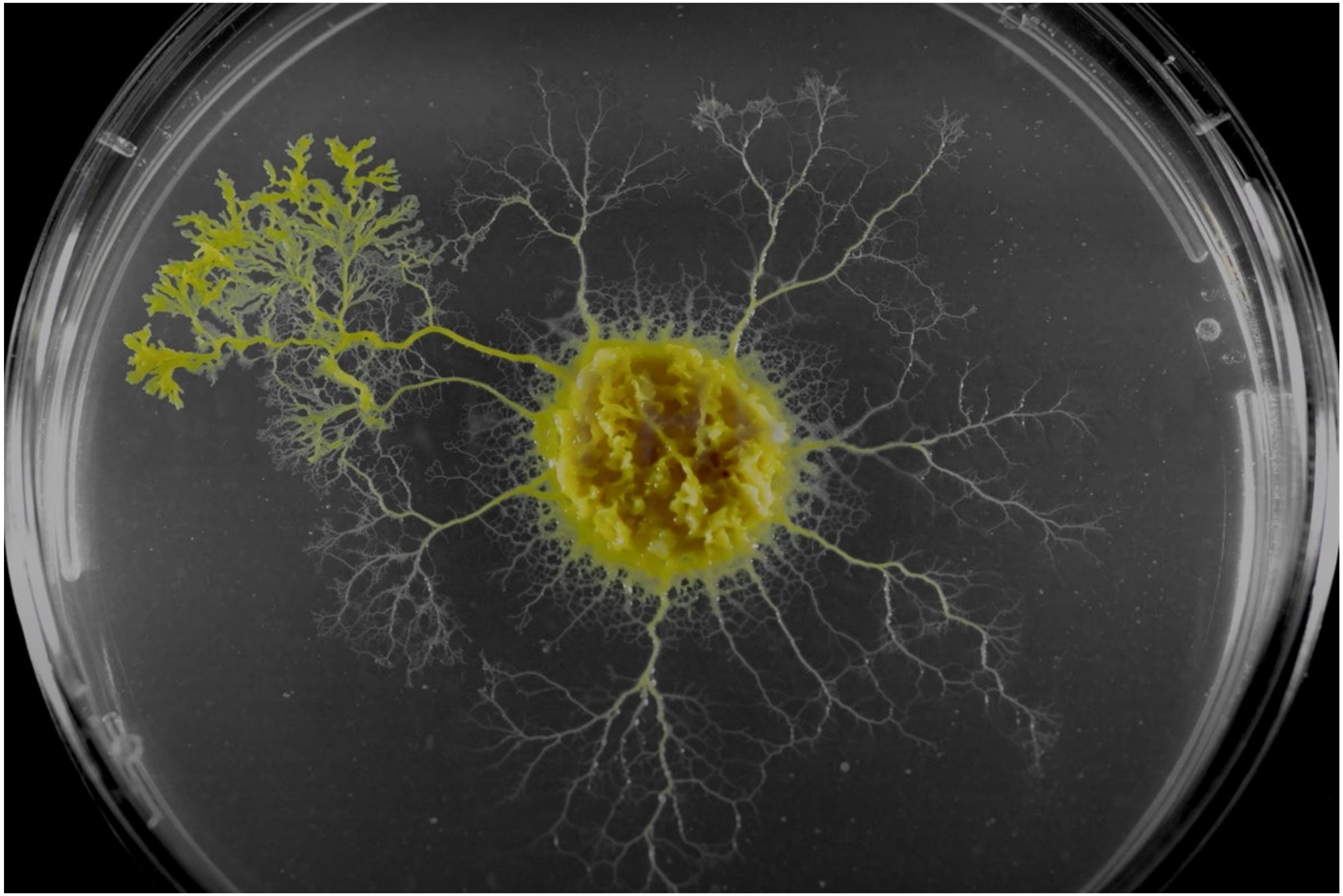

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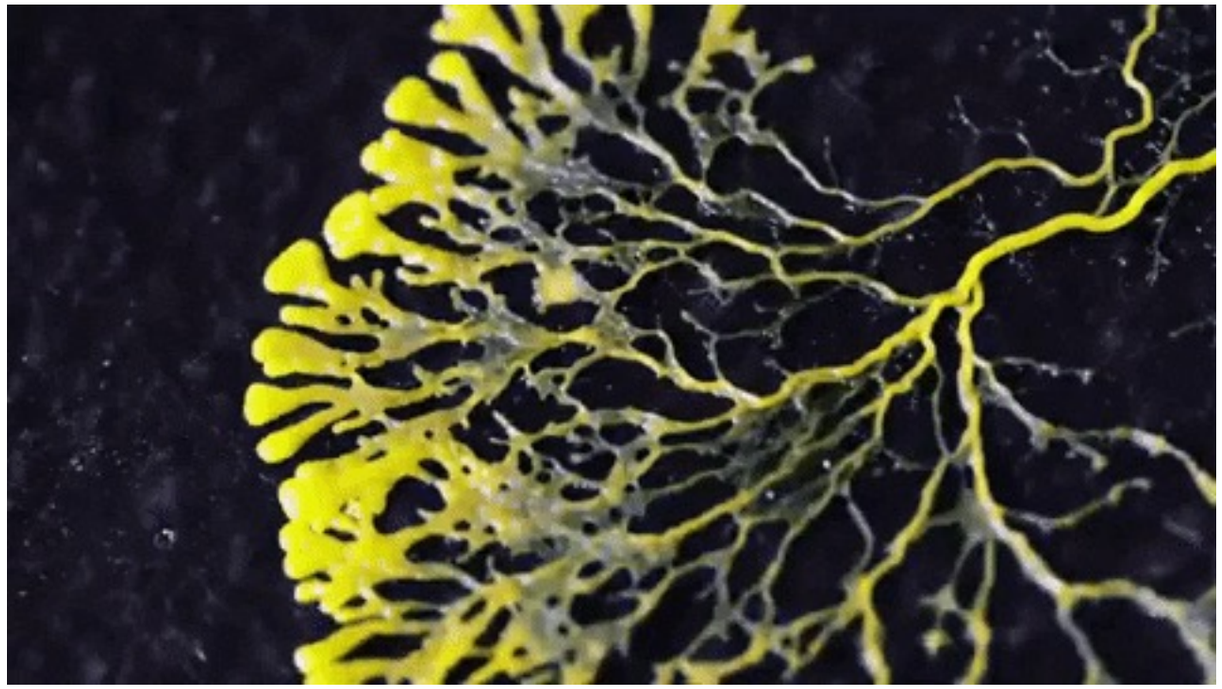

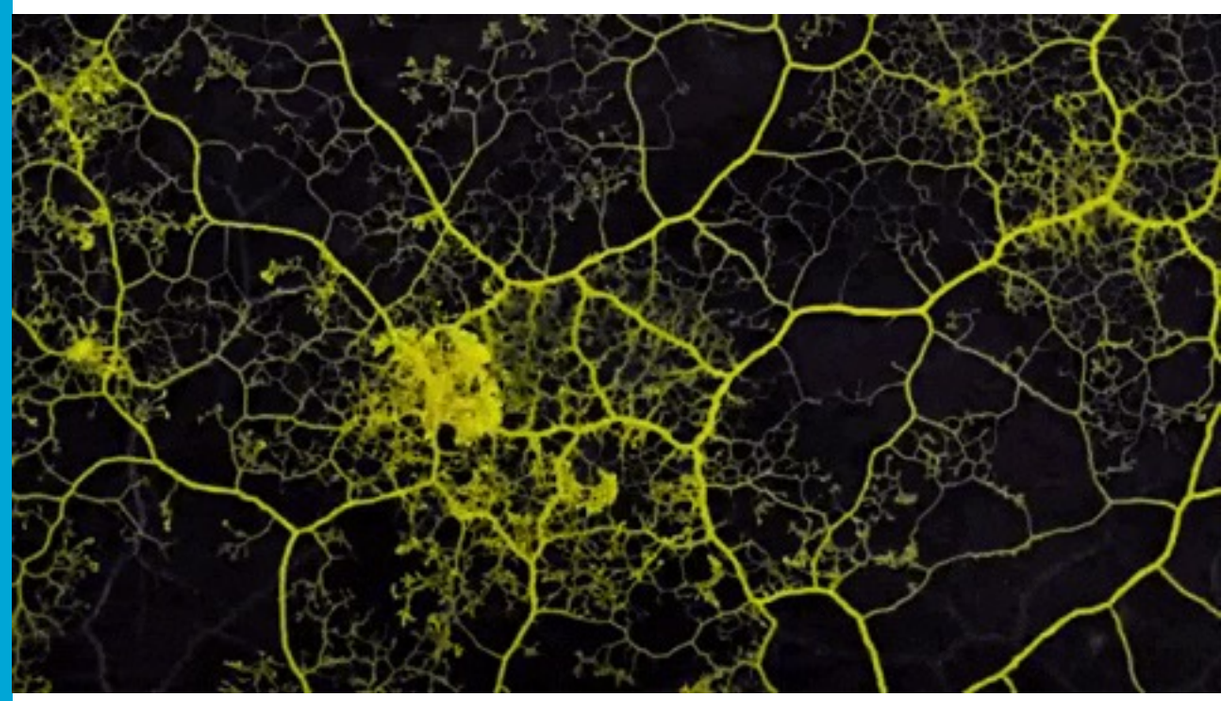

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https://gemachado.medium.com/is-there-intelligent-life-on-chatbots-59be6af8596c

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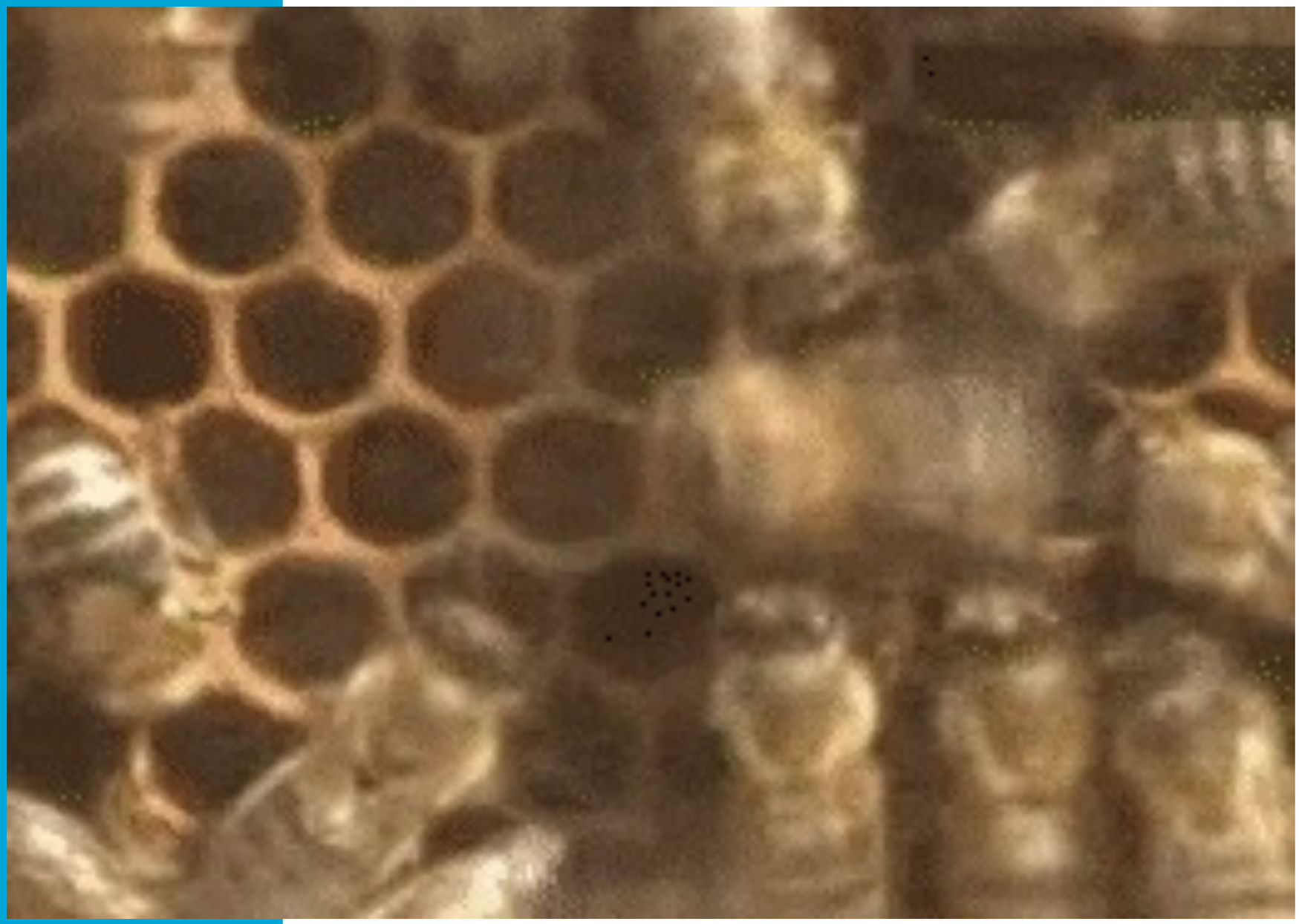

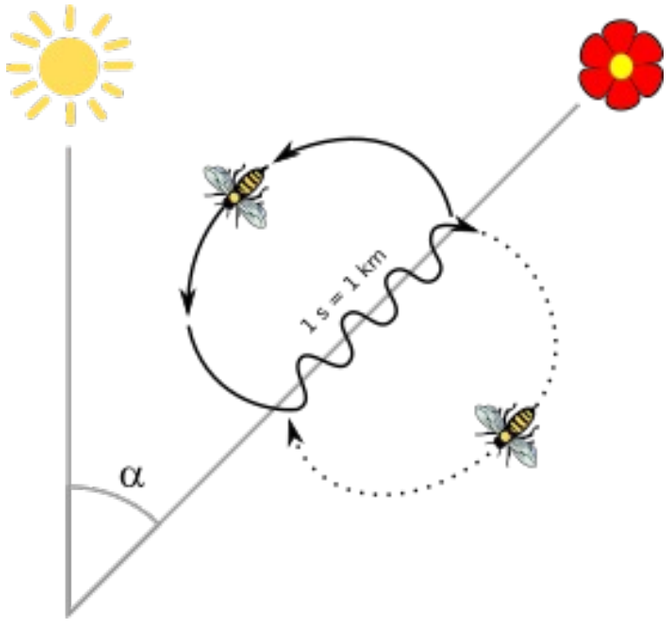

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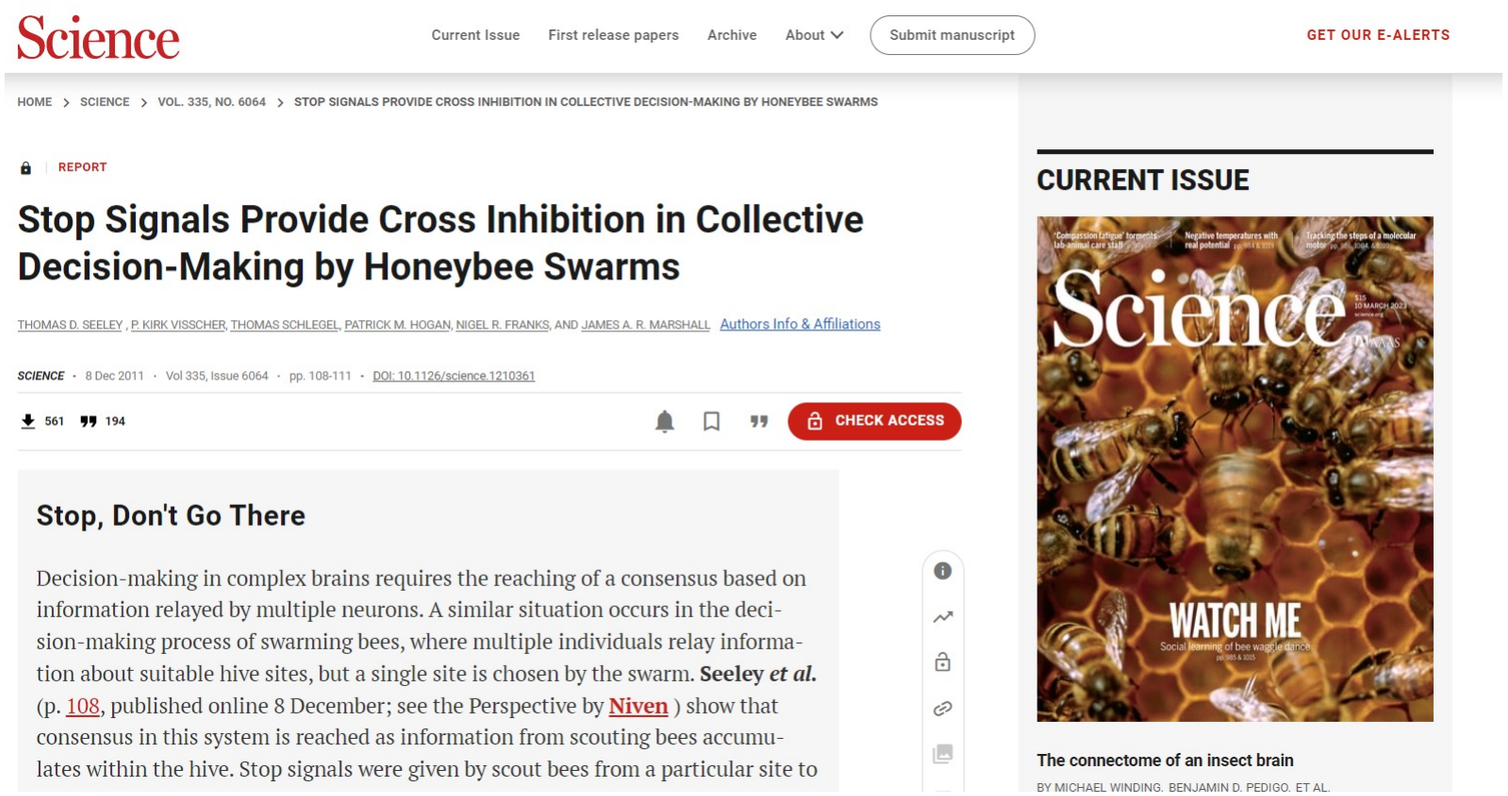
Swarm Intelligence



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HOME > SCIENCE > VOL. 335, NO. 6064 > STOP SIGNALS PROVIDE CROSS INHIBITION IN COLLECTIVE DECISION-MAKING BY HONEYBEE SWARMS

REPORT

Stop Signals Provide Cross Inhibition in Collective Decision-Making by Honeybee Swarms

THOMAS D. SEELEY, P. KIRK VISSCHER, THOMAS SCHLEGEL, PATRICK M. HOGAN, NIGEL R. FRANKS, AND JAMES A. R. MARSHALL Authors Info & Affiliations

SCIENCE • 8 Dec 2011 • Vol 335, Issue 6064 • pp. 108-111 • DOI: 10.1126/science.1210361

561 194 CHECK ACCESS

Stop, Don't Go There

Decision-making in complex brains requires the reaching of a consensus based on information relayed by multiple neurons. A similar situation occurs in the decision-making process of swarming bees, where multiple individuals relay information about suitable hive sites, but a single site is chosen by the swarm. **Seeley *et al.*** (p. 108, published online 8 December; see the Perspective by Niven) show that consensus in this system is reached as information from scouting bees accumulates within the hive. Stop signals were given by scout bees from a particular site to

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The connectome of an insect brain

BY MICHAEL WINDING, BENJAMIN D. PEDIGO, ET AL.

"Decision-making in complex brains requires the reaching of a consensus based on information relayed by multiple neurons. A similar situation occurs in the decision-making process of swarming bees, where multiple individuals relay information about suitable hive sites, but a single site is chosen by the swarm."

https://www.science.org/doi/10.1126/science.1210361

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Swarm Intelligence

- The whole > the sum of the parts
- Complex tasks can be tackled only through cooperation of the multitude
  - **The intelligence is in the group**
- The outcome of the effort comes at a group level
  - The single parts may not be aware of the bigger goal
  - There is no leader or centralized coordination
  - Interaction/coordination happens at a microlevel
    - Direct: through senses
    - Indirect: through changes in the environment
- Complex systems:
  - Self-organization
  - Emergent properties
  - No central planning

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Swarm Intelligence

Can we mimic the emergent behavior of swarms of animals to create intelligent algorithms?

- Computational swarm intelligence:
  - Algorithms that define simple individual behavior modulated by simple interactions between individuals
  - Emergent group behavior
- Focus on the individual actors, not on the whole system (which might be too complex to model)
- We will focus on one illustrative method: Ant Colony Optimization (ACO), but there are many others!

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Swarm Intelligence

Can we mimic the emergent behavior of swarms of animals to create intelligent algorithms?

- Challenges:
  - Find individual behavioral rules that result in desired swarm behavior (reverse engineering).
  - Make sure the emergent behavior is stable.

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Ant Colony Optimization (ACO)

Ant Colonies

- Foraging behavior (act of gathering wild food resources) is based on indirect communication mediated by pheromones
- A pheromone is a secreted or excreted chemical factor which enables ants to effectively explore the environment

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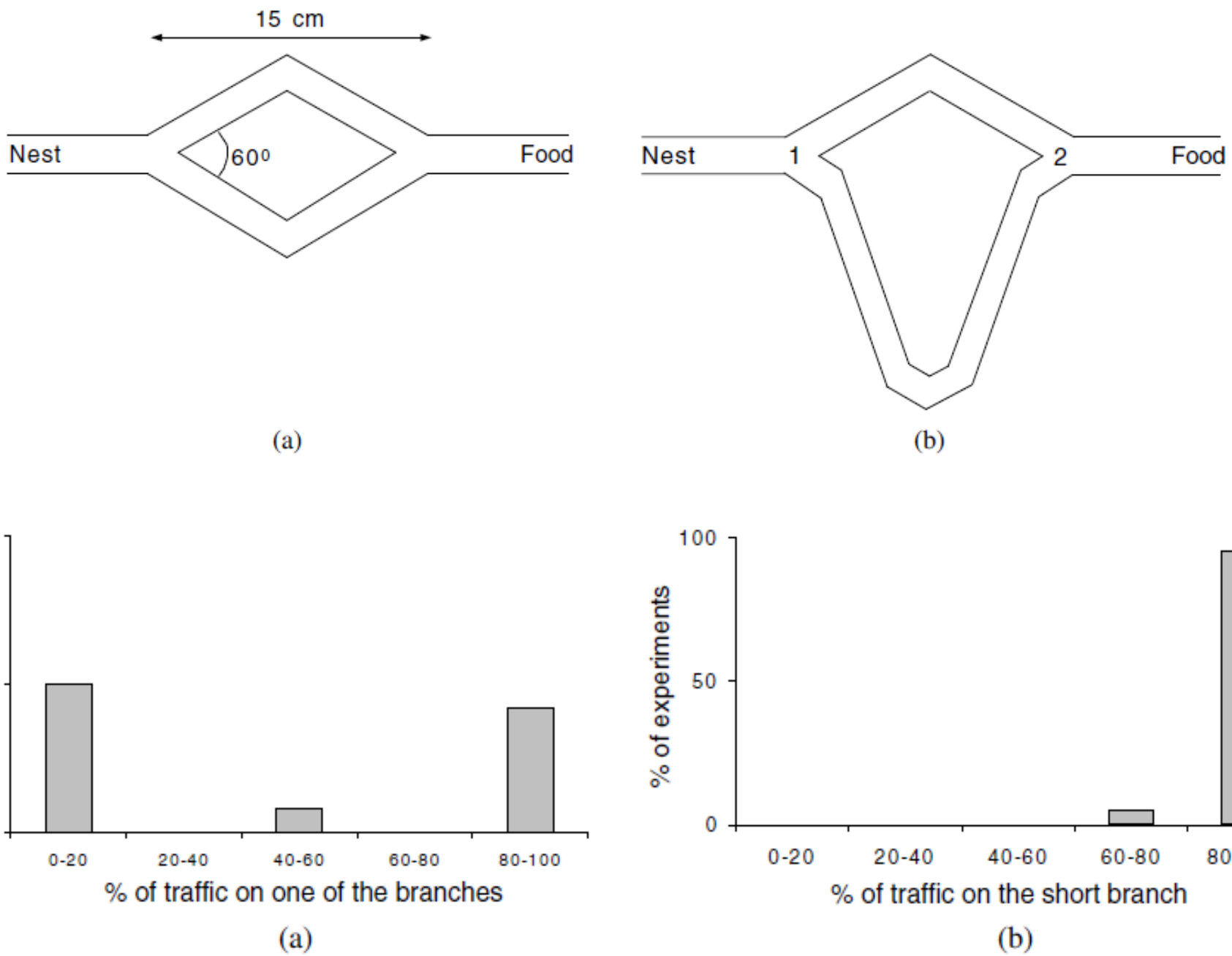
Stigmergy

- Indirect communication mediated by modifications of environmental states which are only locally accessible by the communicating agents
- For example, some ants leave a chemical (pheromone) trail behind to trace the path. The chemical decays over time.
- This allows other ants to find the path between the food and the nest. It also allows ants to find the shortest path among alternative paths.

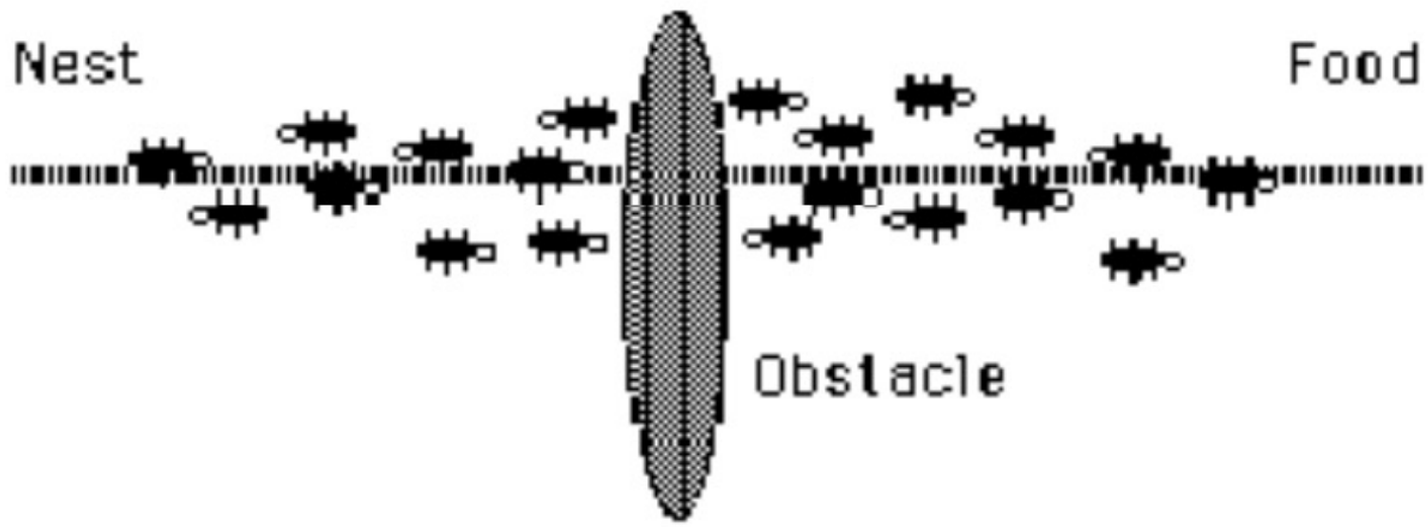
Ant Colonies

- Double bridge experiment (Goss et al, 1989):

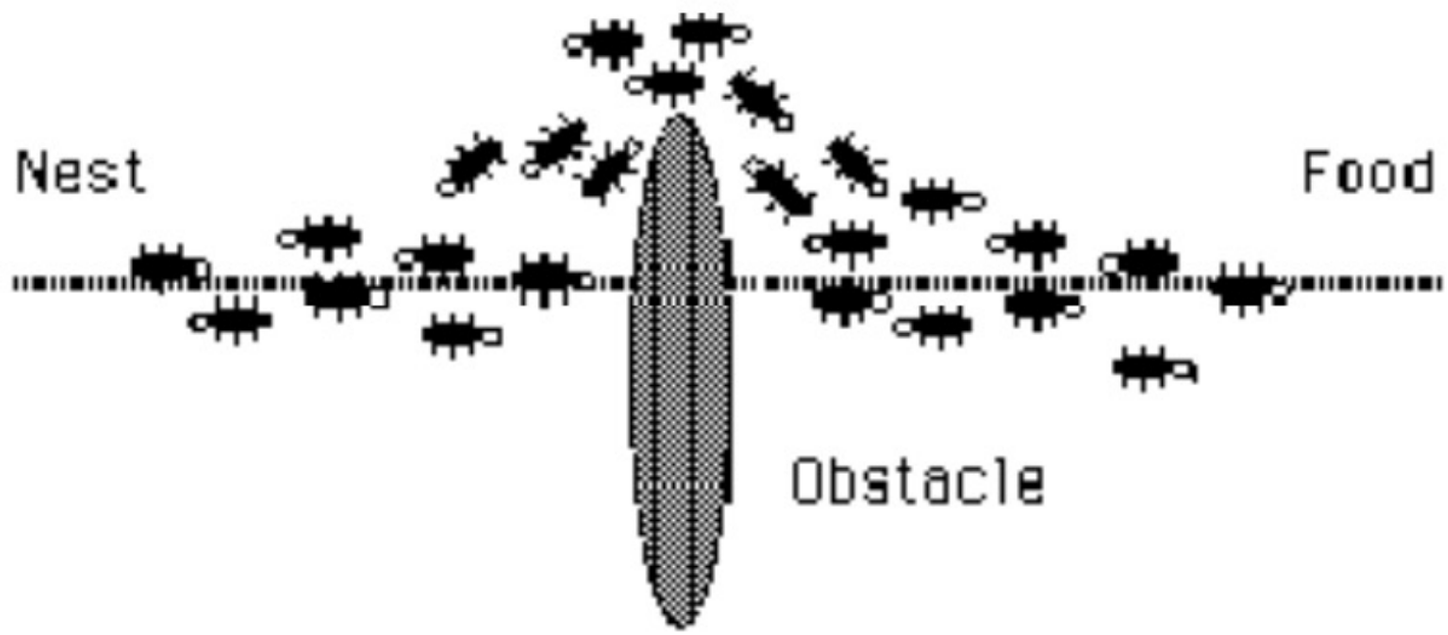


Goss, S., Aron, S., Deneubourg, J. L., & Pasteels, J. M. (1989). Self-organized shortcuts in the Argentine ant. Naturwissenschaften, 76, 579–581.

Nest
Food

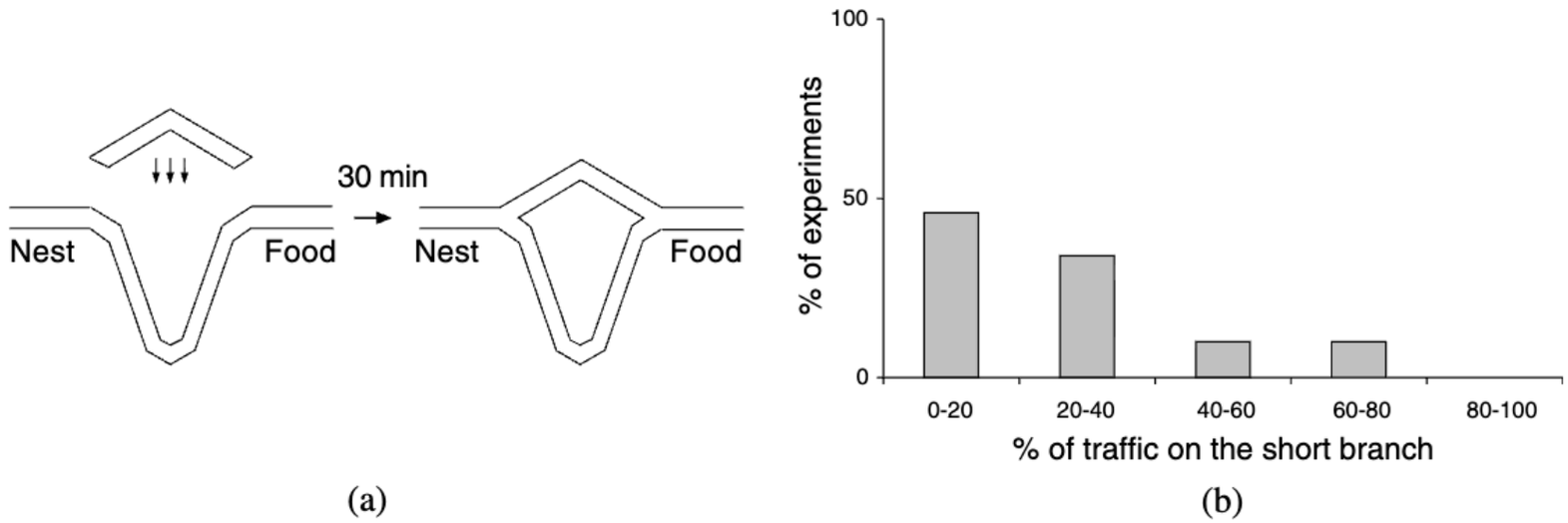

Nest
Food
Obstacle

Nest
Food
Obstacle


Nest
Food
Obstacle

Ant Colonies

Dreyer, T., Haluts, A., Korman, A., Gov, N., Fonio, E., & Feinerman, O. (2024). Comparing cooperative geometric puzzle solving in ants versus humans. *Proceedings of the National Academy of Sciences*, *122*(1), e2414274121. https://www.pnas.org/doi/abs/10.1073/pnas.2414274121



Dorigo, Marco, Mauro Birattari, and Thomas Stutzle. "Ant colony optimization." *IEEE computational intelligence magazine* 1.4 (2006): 28-39.

Ant Colony Optimization

- **Artificial ants** solve optimization problems, inspired by how real ants communicate (indirectly) via pheromone

- For example, the task for “artificial ants” can be to **find the shortest way to some goal**
  - Initially all links have the same (non-zero) probability to be explored
  - When exploring different links, ants release a quantity of pheromone τ, proportional to the length of the link (calculated at the end of course)
  - The probability that an ant choses a specific path depends on the amount of pheromone that was deposited on it
  - The process is **stochastic** → Near-optimal solutions

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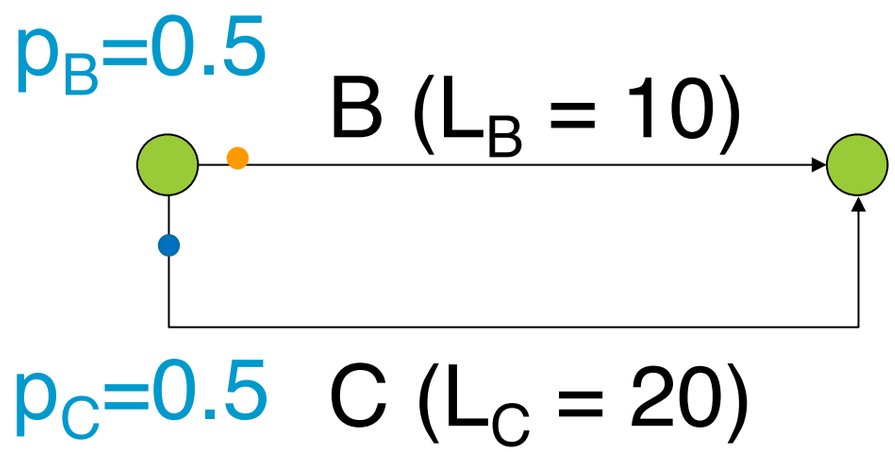
Ant Colony Optimization: How?



Two paths to the goal: B and C

Ant 1 chooses B: distance travelled: $L_B = 10$

Ant 2 chooses C: distance travelled $L_C = 20$

The shorter the distance, the more pheromone is released on the link

Assuming $\boldsymbol{Q} = \mathbf{100}$

Pheromone left on the i-th link by ant k

Constant

$$\Delta\tau_i^k = \frac{Q}{L_i}$$

link length

$$\Delta\tau_B^1 = \frac{100}{10} = 10$$

$$\Delta\tau_C^2 = \frac{100}{20} = 5$$

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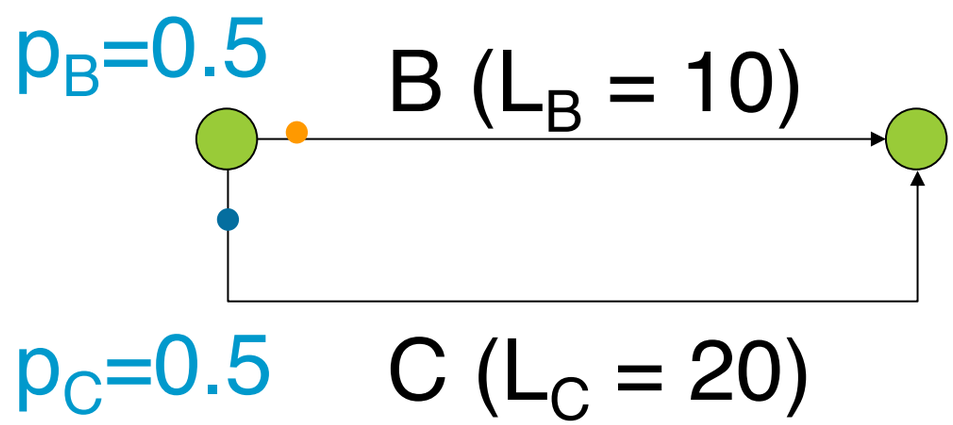
Ant Colony Optimization

Probabilities to chose the path are updated according to the amount of pheromone



$$p_B = \frac{\tau_B}{\tau_B + \tau_C} = \frac{10}{15} = \frac{2}{3}$$

$$p_C = \frac{\tau_C}{\tau_B + \tau_C} = \frac{5}{15} = \frac{1}{3}$$

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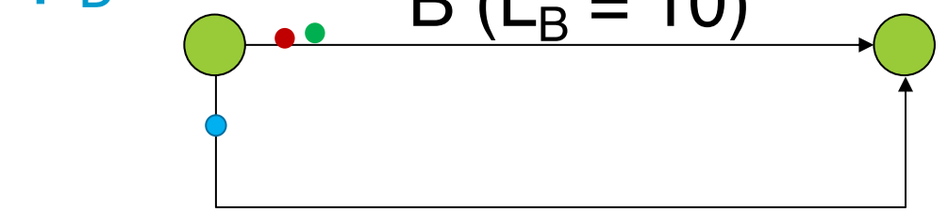
Ant Colony Optimization



Ant 3 chooses B: L_B = 10, $\Delta\tau_B^3$ = 100 / 10 = 10

Ant 4 chooses C: L_c = 20, $\Delta\tau_C^4$ = 100 / 20 = 5

Ant 5 chooses B: L_B = 10, $\Delta\tau_B^5$= 100 / 10 = 10

Assuming total pheromone on link i

$$\tau_i = \sum_k \Delta\tau_i^k$$

This is a simplification! We will see more elaborate methods later in the lecture

$$\tau_B = \tau_B^3 + \tau_B^5 = 10 + 10 = 20$$

$$\tau_C = \tau_B^4 = 5$$

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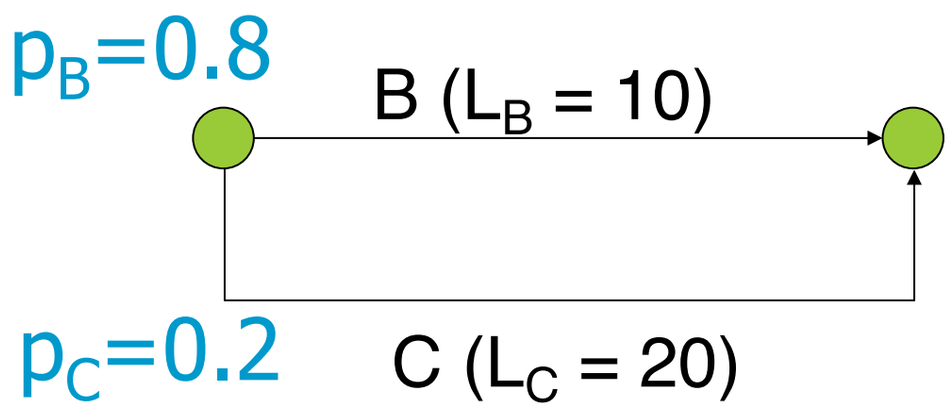
Ant Colony Optimization



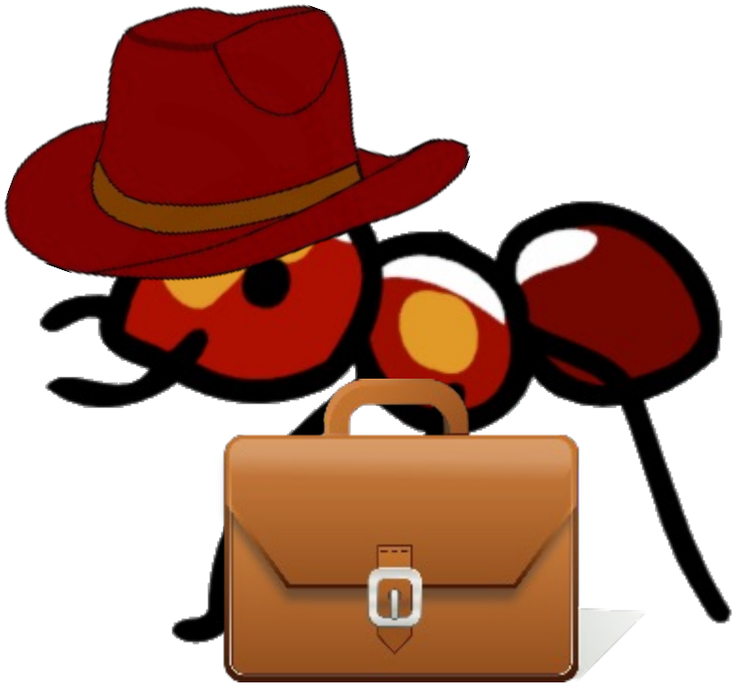
$$p_B = \frac{\tau_B}{\tau_B + \tau_C} = \frac{20}{25} = \frac{4}{5}$$

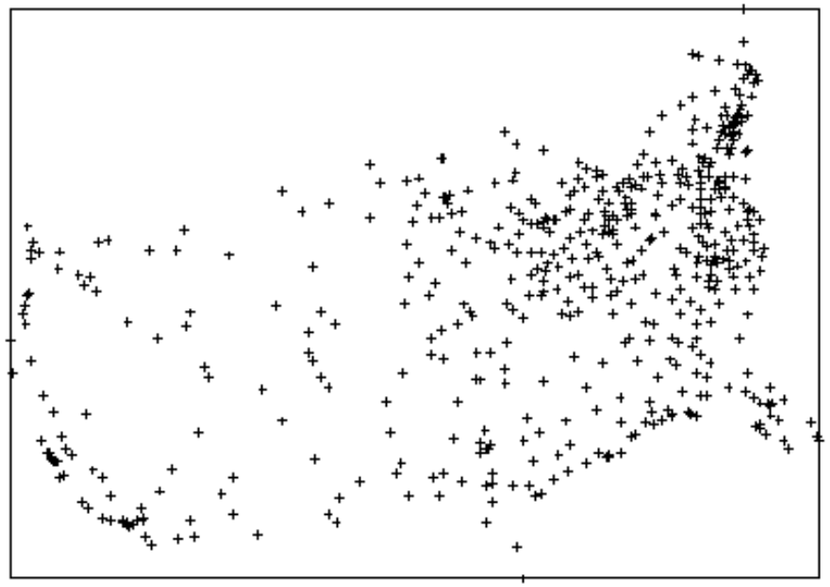
$$p_C = \frac{\tau_C}{\tau_B + \tau_C} = \frac{5}{25} = \frac{1}{5}$$

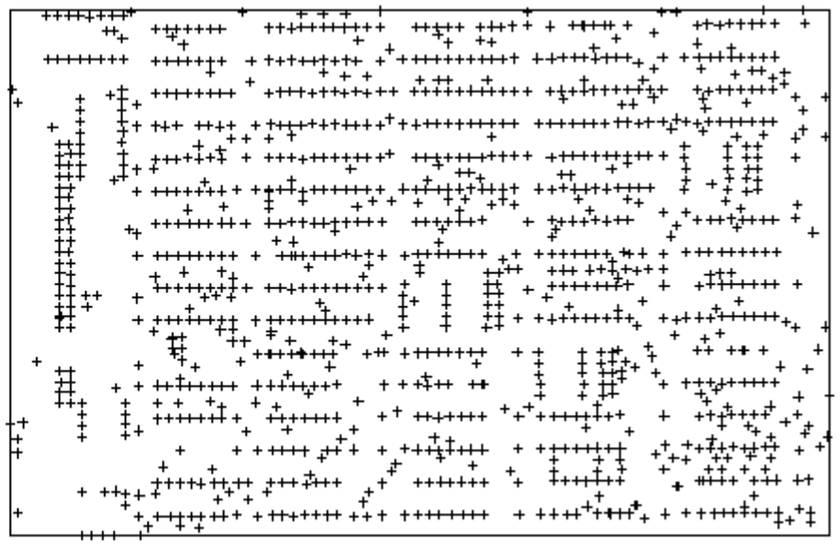
The Travelling Salesman Problem (TSP)

Find the shortest path that allows visiting each city exactly once (many applications, e.g. logistics)

- The solution requires a number of computational steps that grows faster than the number of cities
- 15 cities = $\frac{(n-1)!}{2} = 43\,589\,145\,600$ combinations







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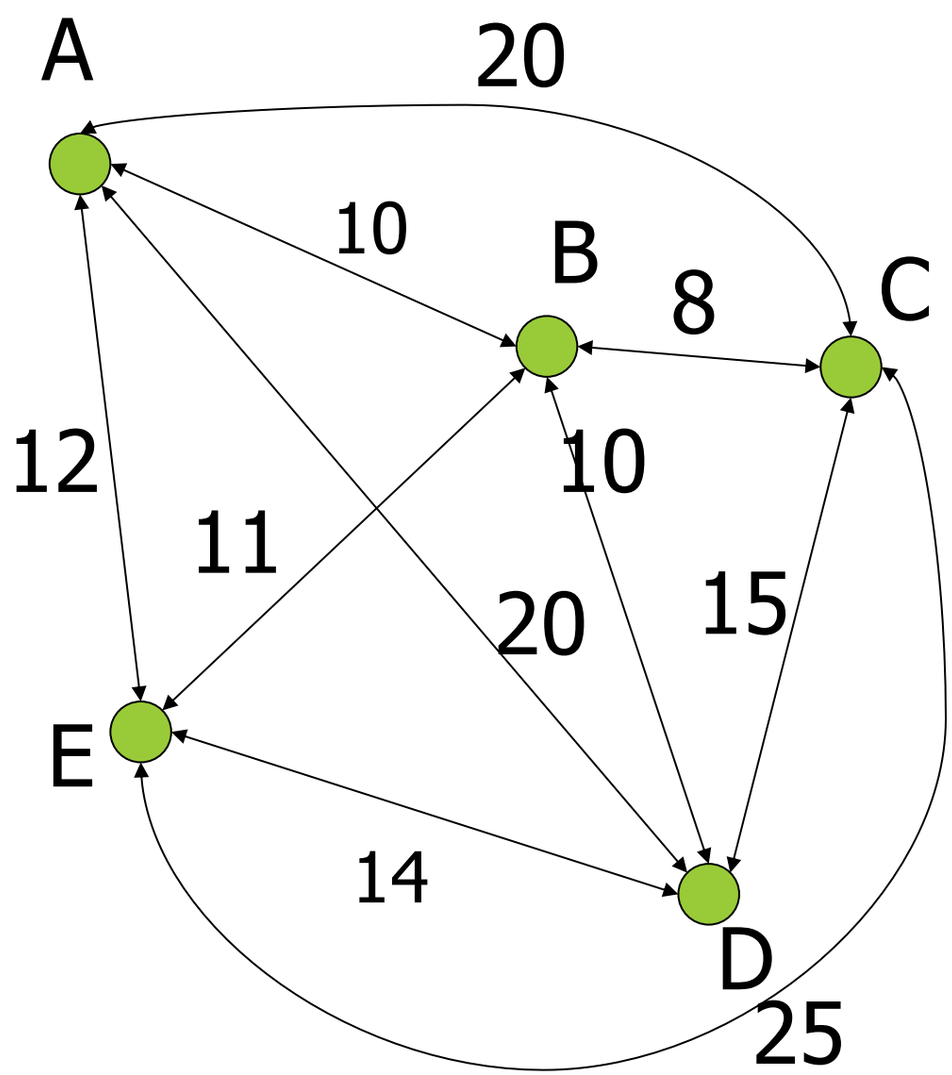
Travelling Sales Ants

Employ artificial ants to solve TSP



Assumptions:

- Ants have some memory (they know which cities were visited before)
- Ants are not completely blind (they know to which city a given link goes)
- Time is discrete
- Starting points are randomly assigned

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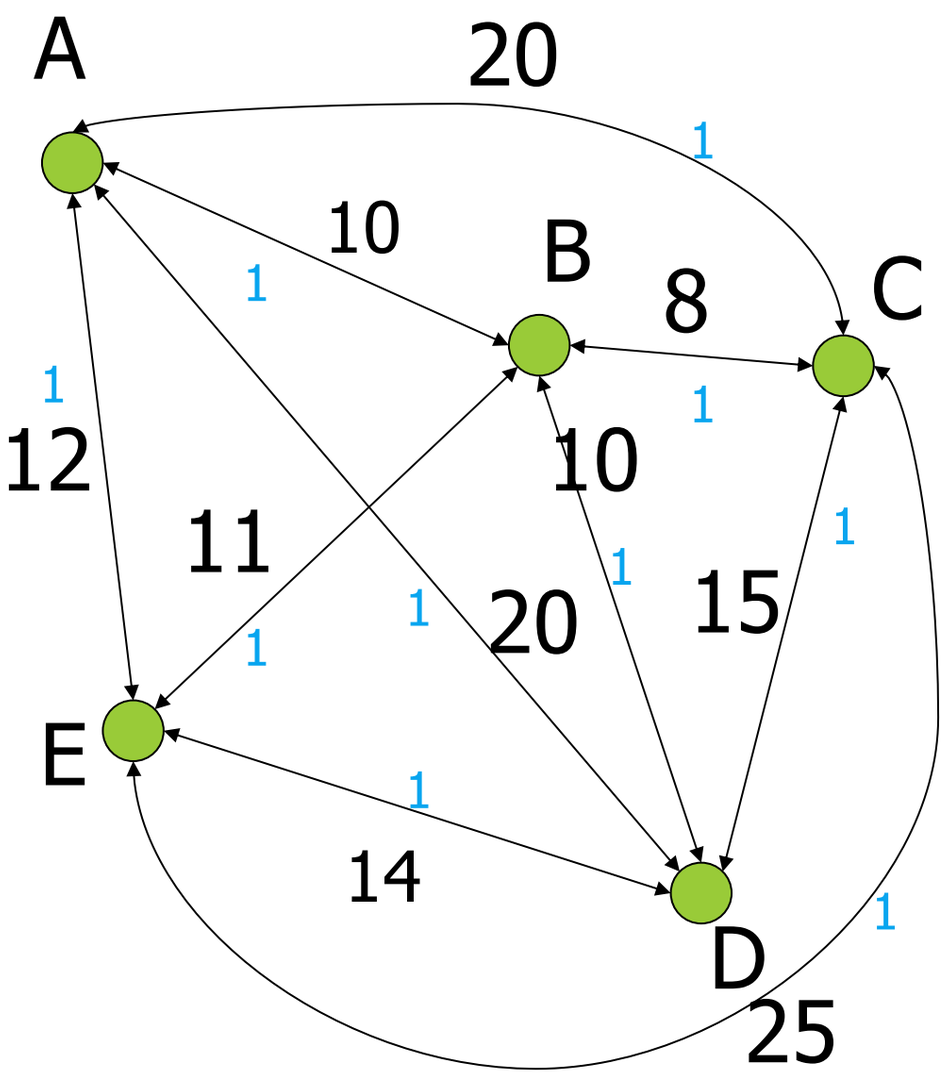
Ant Colony Optimization for TSP



Ant 1: Starting in A

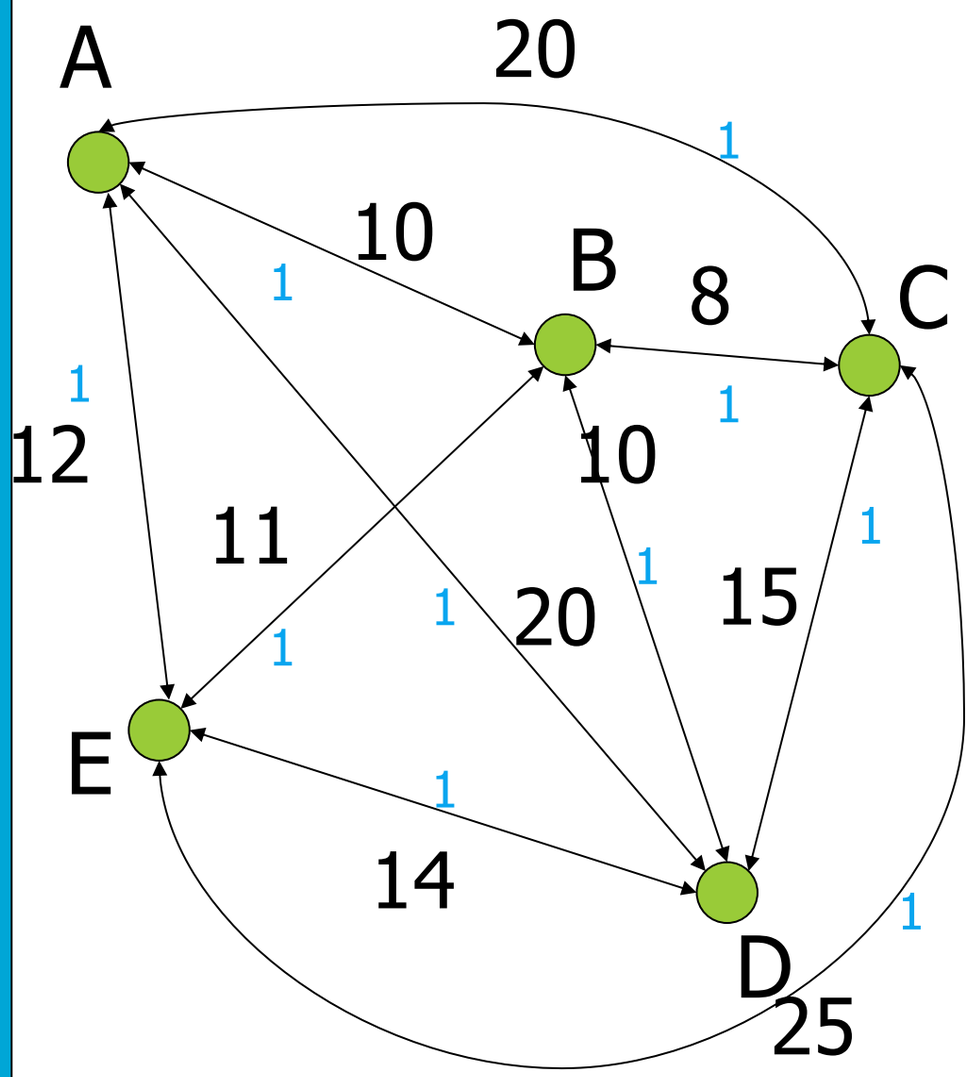
A→B→C → E→D→A

10
8
25
14
20

$L^1 =$ 77

*little blue numbers on links represent the quantity of pheromone

Ant Colony Optimization for TSP



Ant 1: Starting in A

A→B→ C→ E→D→A

10
8
25
14
20

$L^1 = 77$

Ant 2: Starting in C

C→B→E→A→D→C

8
11
12
20
15

$L^2 = 66$

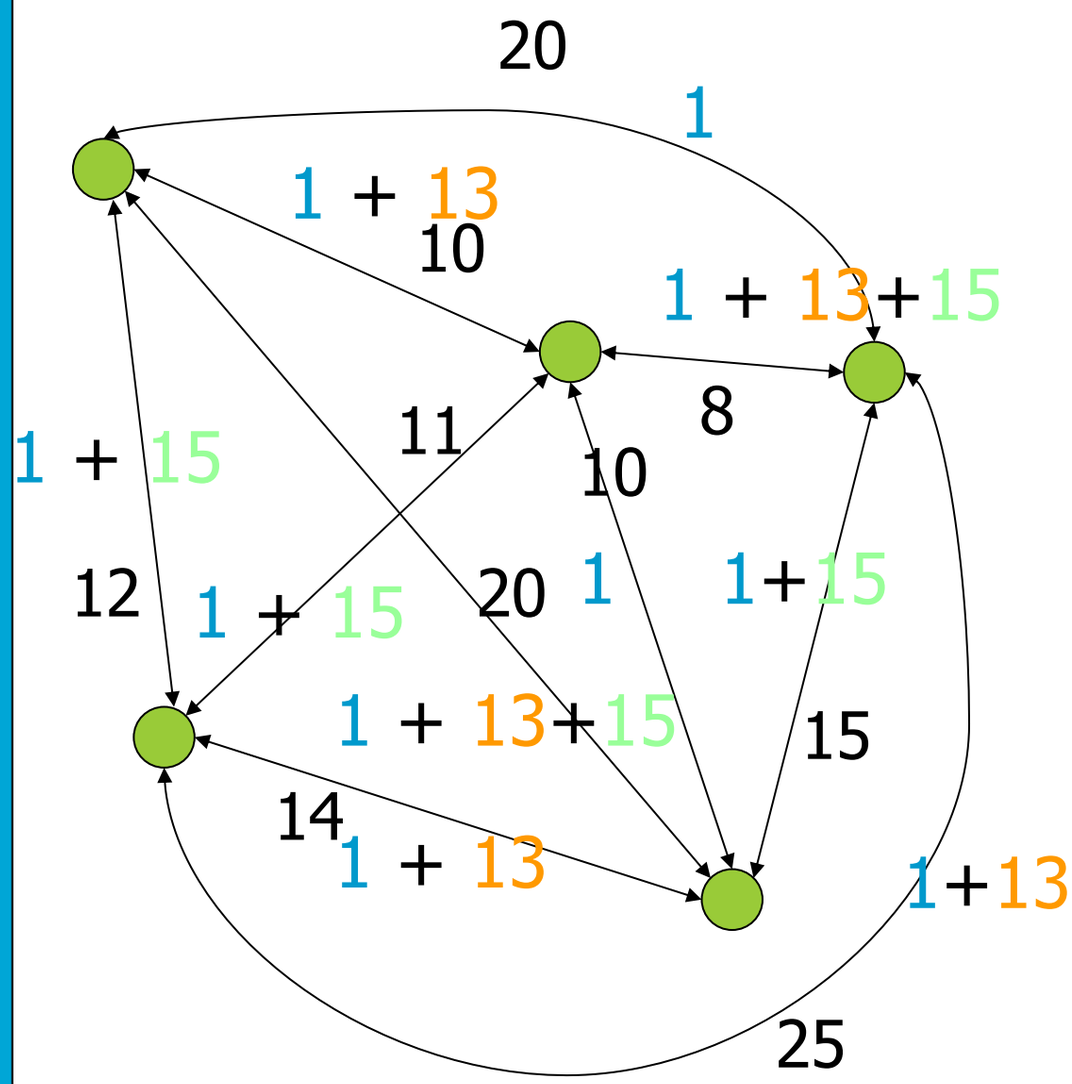
Shortest route gets more pheromone:

$$\Delta\tau^k = Q \times \frac{1}{L^k}$$

Assuming: Q = 1000

$$\Delta\tau^1 = \frac{1000}{77} = 12.99 \qquad \Delta\tau^2 = \frac{1000}{66} = 15.15$$

Ant Colony Optimization for TSP

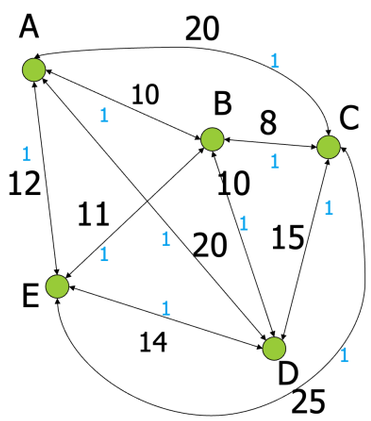


Ant 1: 12.99 ≅ 13

Ant 2: 15.15 ≅ 15

$$\tau_{ij} = \tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^{k}$$

Ref:



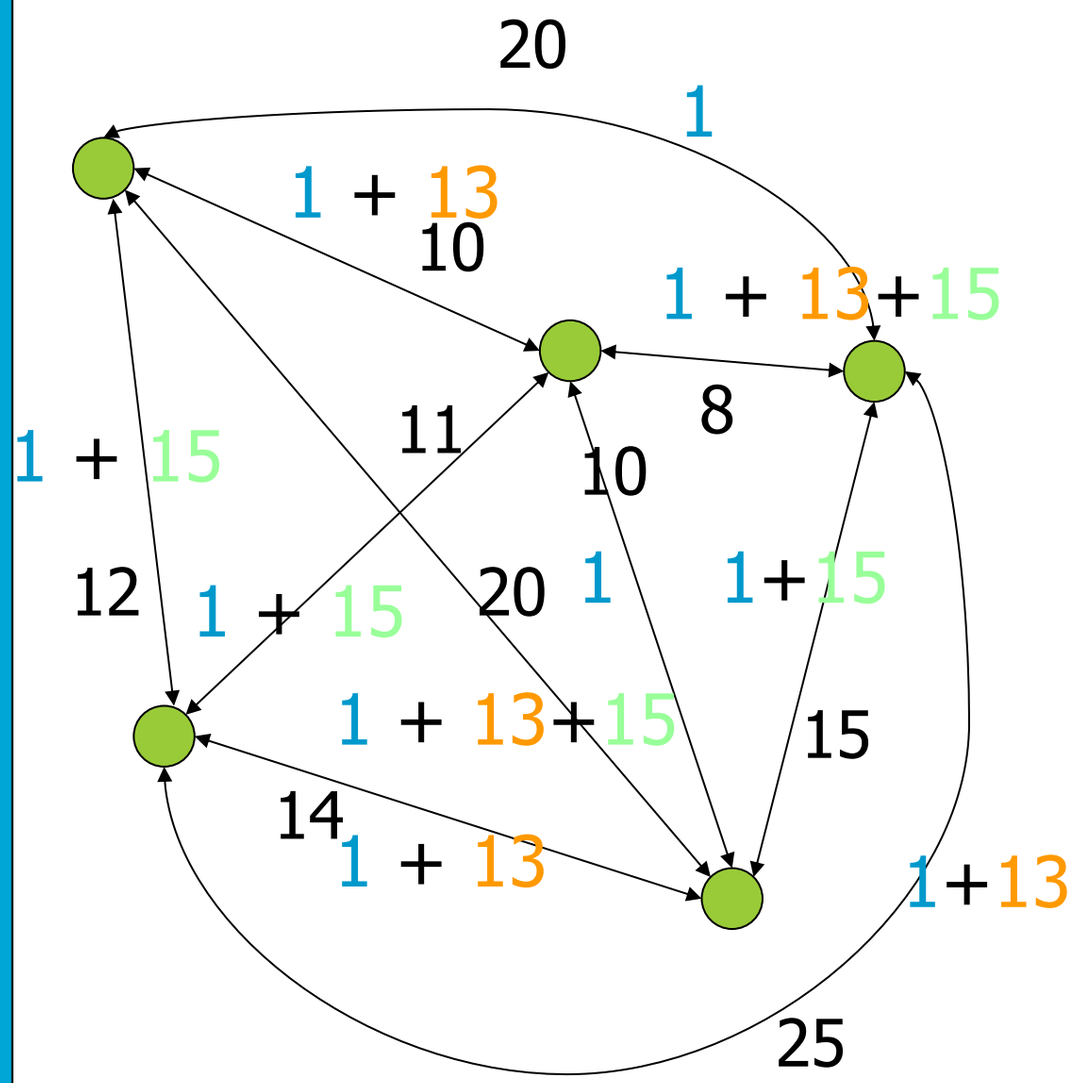
Ant 1: Starting in A

A→B→ C→ E→D→A

Ant 2: Starting in C

C→B→E→A→D→C

Ant Colony Optimization for TSP



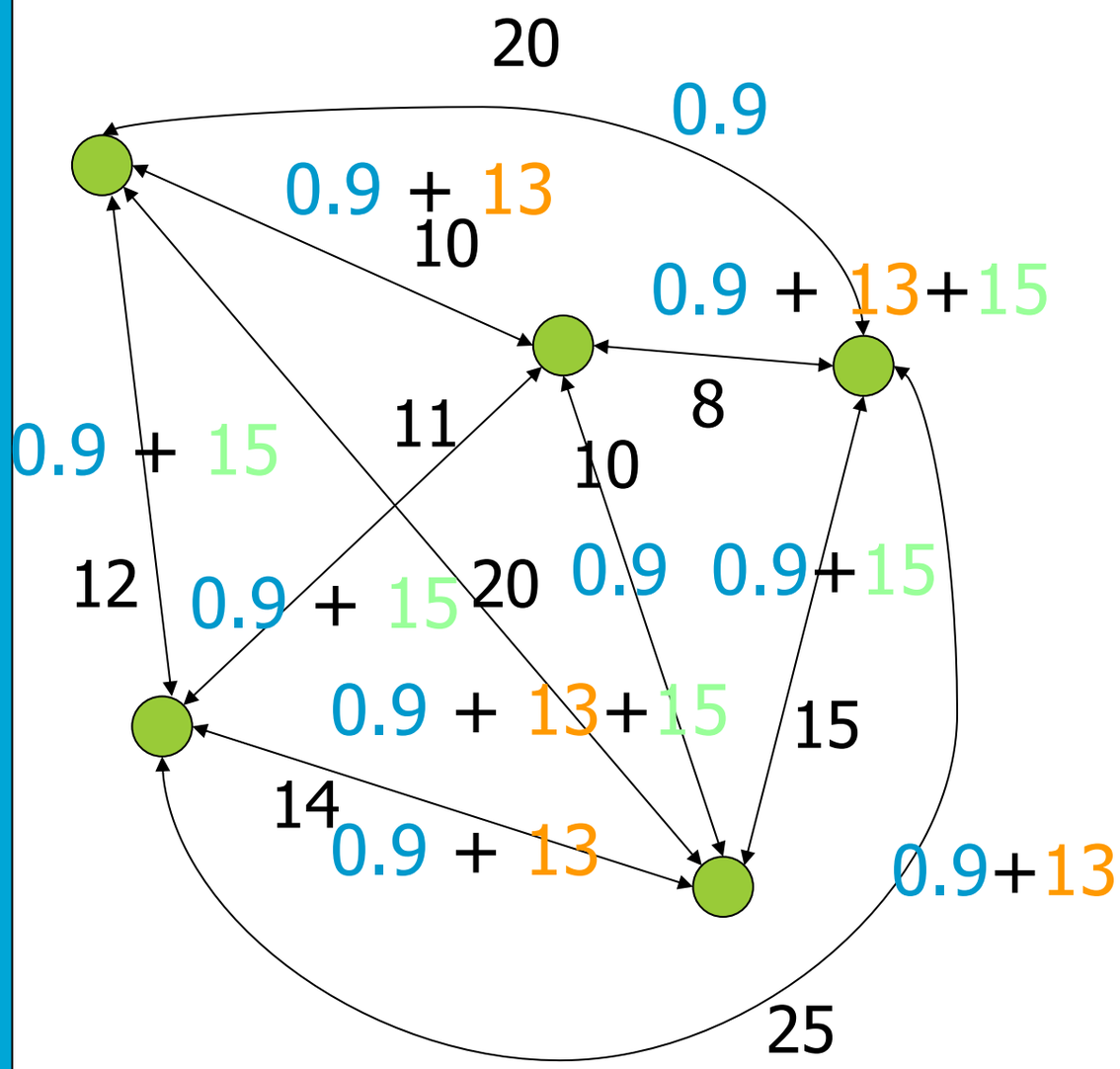
Ant 1: 12.99 ≅ 13

Ant 2: 15.15 ≅ 15

$$\tau_{ij} = \tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^{k}$$

- What if many routes use a link which is not part of a shorter tour?
- What if a short route contains a very long link that initially is less likely to be used?

Ant Colony Optimization for TSP



Ant 1: 12.99 ≅ 13

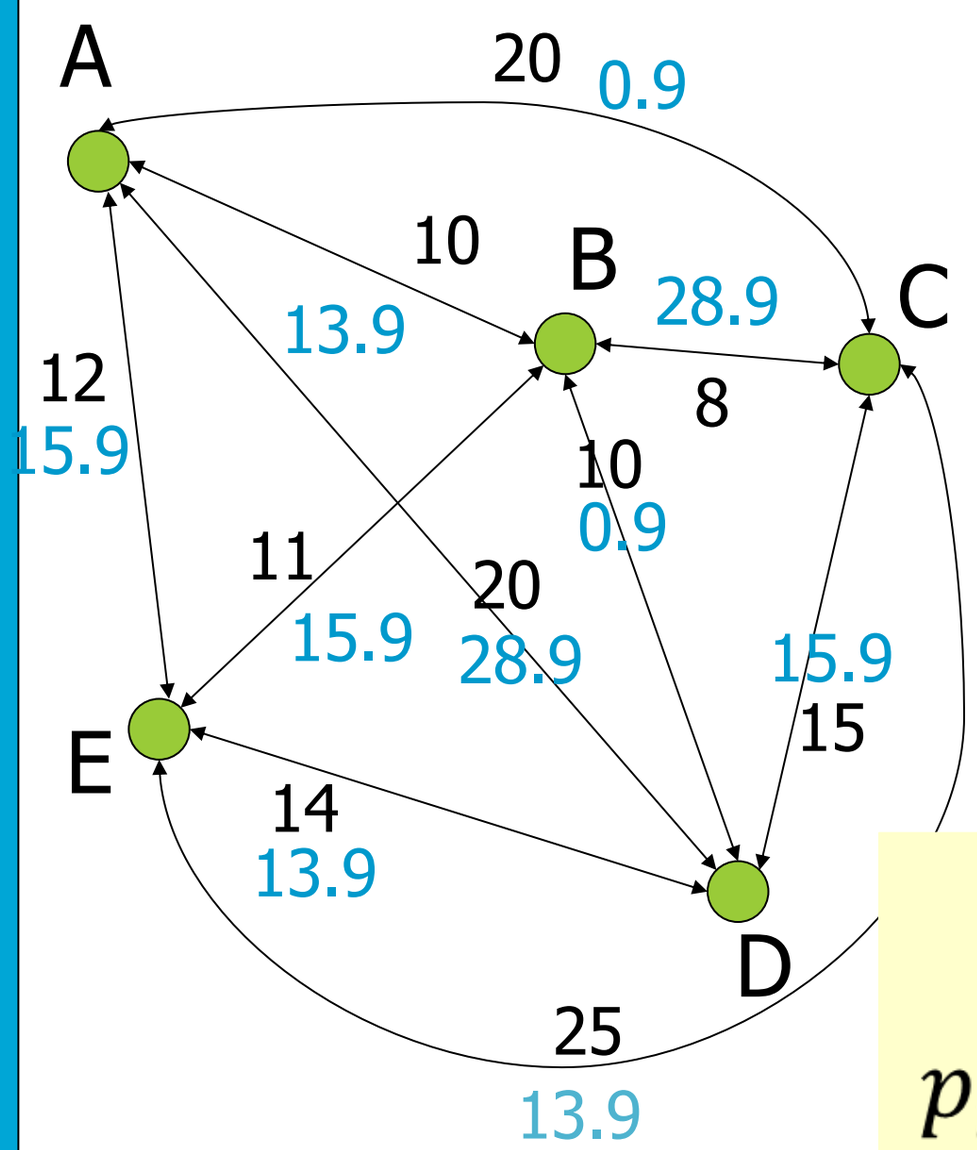
Ant 2: 15.15 ≅ 15

$$\tau_{ij} = (1 - \rho) \cdot \tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^{k}$$

ρ: evaporation constant (here 0.1)

The amount of pheromone currently on path ij corresponds to the amount of pheromone existing on it, decreased by a quantity proportional to the evaporation constant, plus the sum of all the pheromone released on that link by all the ants that passed by it this round.

Ant Colony Optimization for TSP



The probability of choosing for a (previously unseen) path is proportional to the amount of pheromone and inversely proportional to the length of the path

Heuristic (can be another), e.g. Euclidean distance

$$\eta_{ij} = \frac{1}{L_{ij}}$$

Pheromone on path ij

Set of all unvisited cities that can be reached from i

$$p_{ij}^k(t) = \begin{cases} \dfrac{\tau_{ij}(t)^\alpha \eta_{ij}^\beta}{\sum_{m \in C_i^k} \tau_{im}(t)^\alpha \eta_{im}^\beta}, & j \in C_i^k \\ 0, & \text{otherwise} \end{cases}$$

Pheromone on all paths departing from city i

α and β are pre-specified parameters (trail x visibility)

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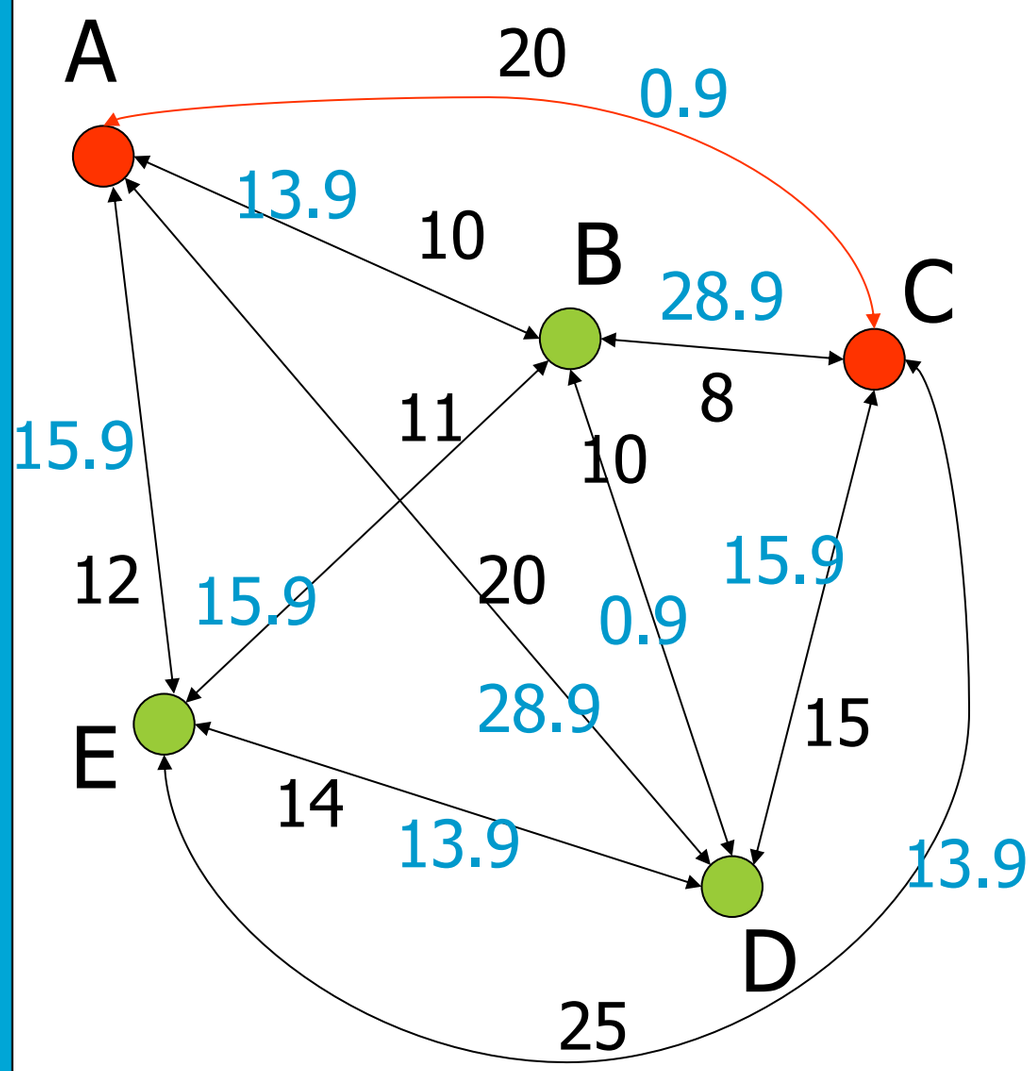
Ant Colony Optimization for TSP



Ant 1 in C (coming from A):

$$p_{CB} = \frac{28.9 * (1/8)^{0.5} \quad (= 10.2177)}{10.2177 + 4.105 + 2.78}$$

$$p_{CD} = \frac{15.9 * (1/15)^{0.5} \quad (= 4.105)}{10.2177 + 4.105 + 2.78}$$

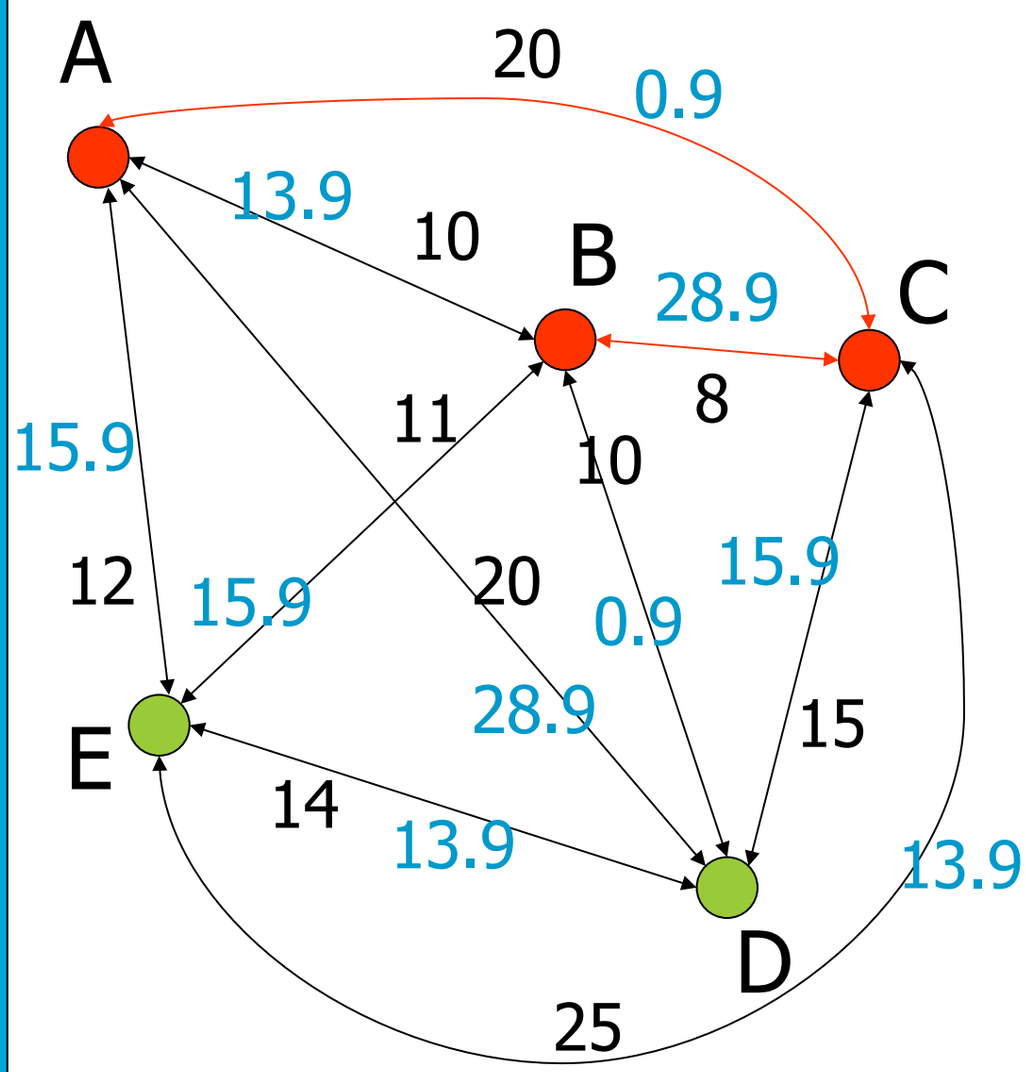
$$p_{CE} = \frac{13.9 * (1/25)^{0.5} \quad (= 2.78)}{10.2177 + 4.105 + 2.78}$$

$$p_{ij}^{k}(t) = \begin{cases} \dfrac{\tau_{ij}(t)^{\alpha}\eta_{ij}^{\beta}}{\sum_{m \in C_i^k} \tau_{im}(t)^{\alpha}\eta_{im}^{\beta}}, & j \in C_i^k \\ 0, & \text{otherwise} \end{cases}$$

Assuming:

$\alpha = 1$, $\beta = 0.5$

Ant Colony Optimization for TSP



$p_{CB} = 0.597$

Let us assume this path was chosen

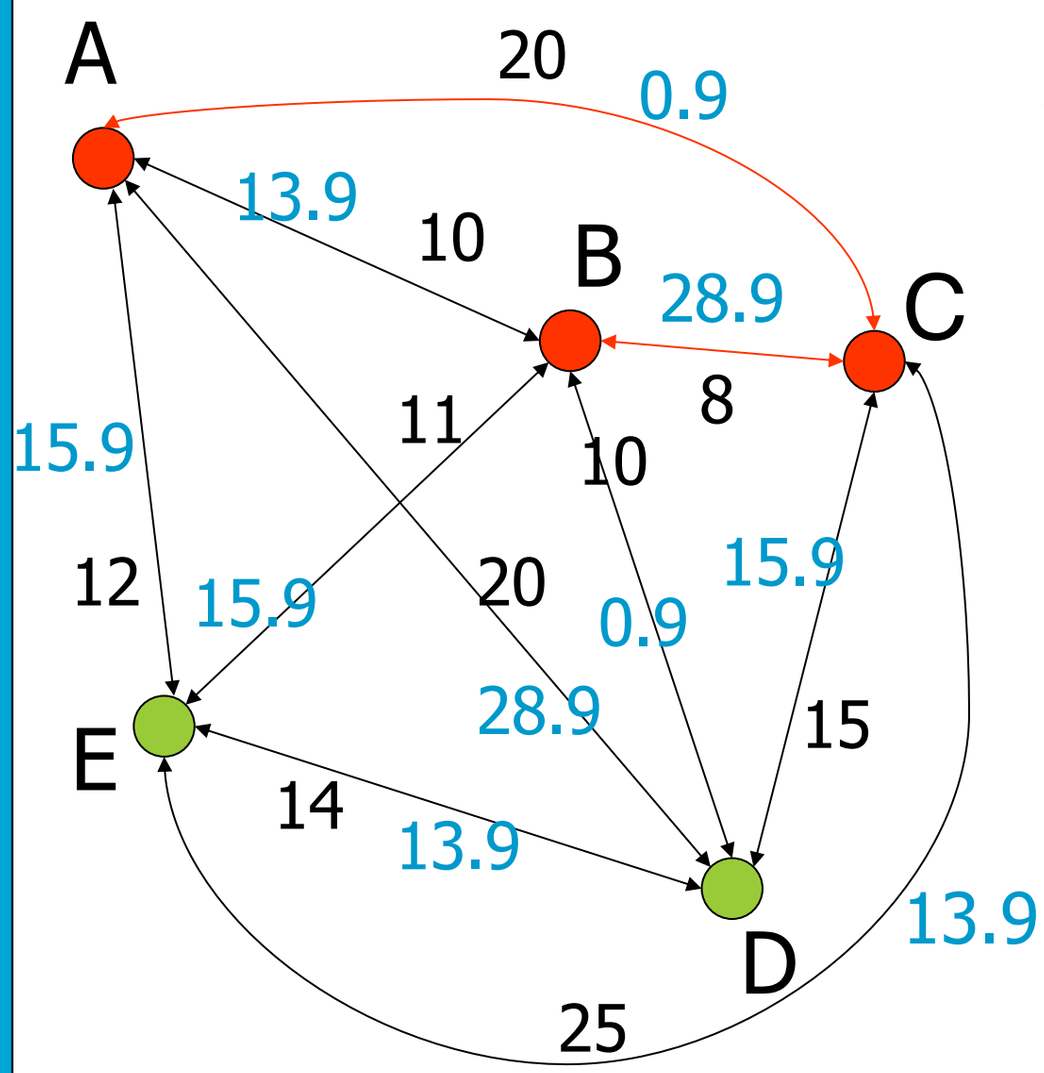
$p_{CD} = 0.240$

$p_{CE} = 0.163$

$$p_{ij}^{k}(t) = \begin{cases} \dfrac{\tau_{ij}(t)^{\alpha}\eta_{ij}^{\beta}}{\sum_{m \in C_i^k} \tau_{im}(t)^{\alpha}\eta_{im}^{\beta}}, & j \in C_i^k \\ 0, & \text{otherwise} \end{cases}$$

$\alpha = 1, \beta = 0.5$

Ant Colony Optimization for TSP



Ant 1 in B (coming from C):

$$p_{BD} = \frac{0.9 * (1/10)^{0.5}}{0.9*(1/10)^{0.5}+15.9 * (1/11)^{0.5}}$$

$= 0.056$

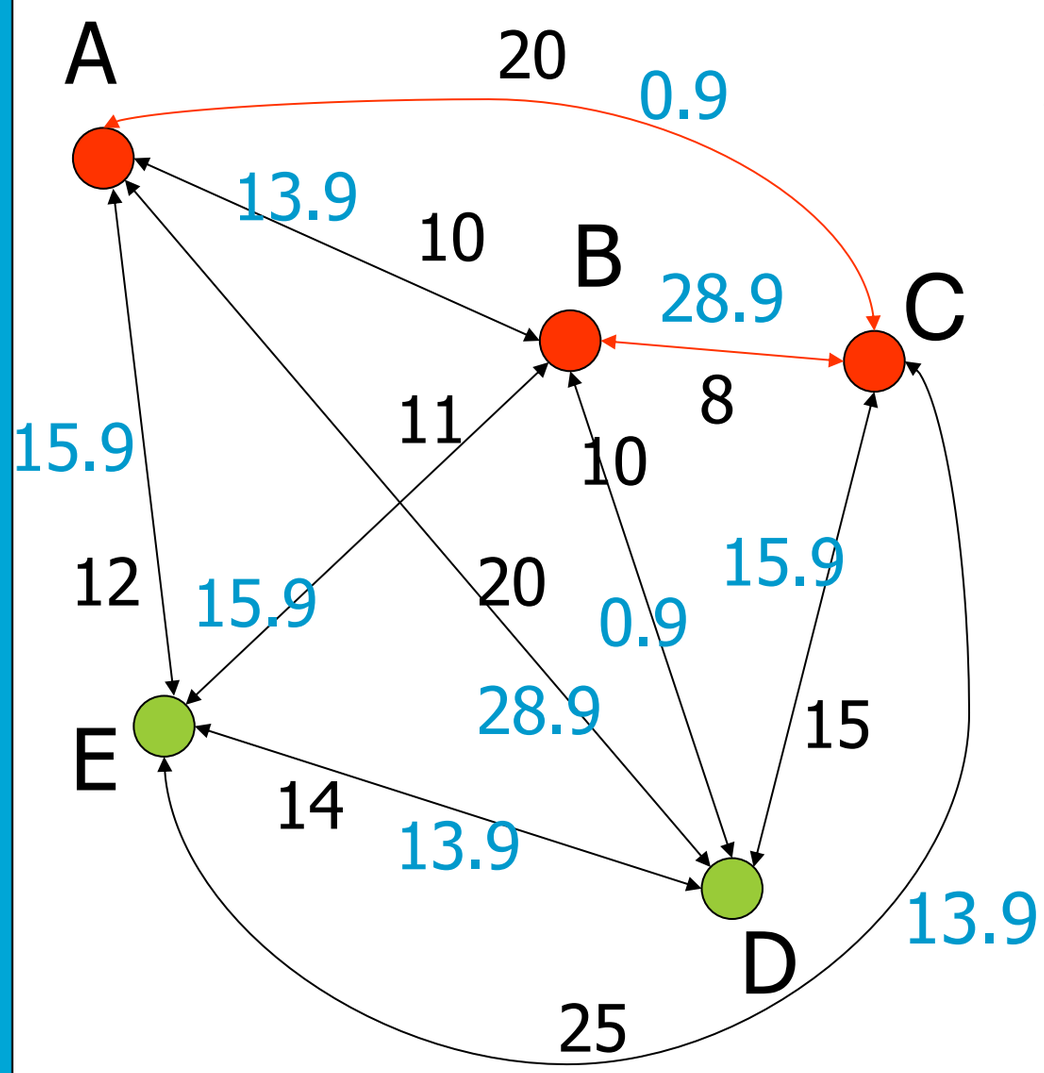
$$p_{BE} = \frac{15.9 * (1/11)^{0.5}}{0.9*(1/10)^{0.5}+15.9 * (1/11)^{0.5}}$$

$= 0.944$

$$p_{ij}^{k}(t) = \begin{cases} \dfrac{\tau_{ij}(t)^{\alpha}\eta_{ij}^{\beta}}{\sum_{m \in C_i^k} \tau_{im}(t)^{\alpha}\eta_{im}^{\beta}}, & j \in C_i^k \\ 0, & \text{otherwise} \end{cases}$$

$\alpha = 1, \beta = 0.5$

Ant Colony Optimization for TSP



Ant 1 in B (coming from C):

$p_{BD} = 0.056$

Let us assume that, even being more unlikely, Ant 1 still chooses to go to node D (stochasticity)

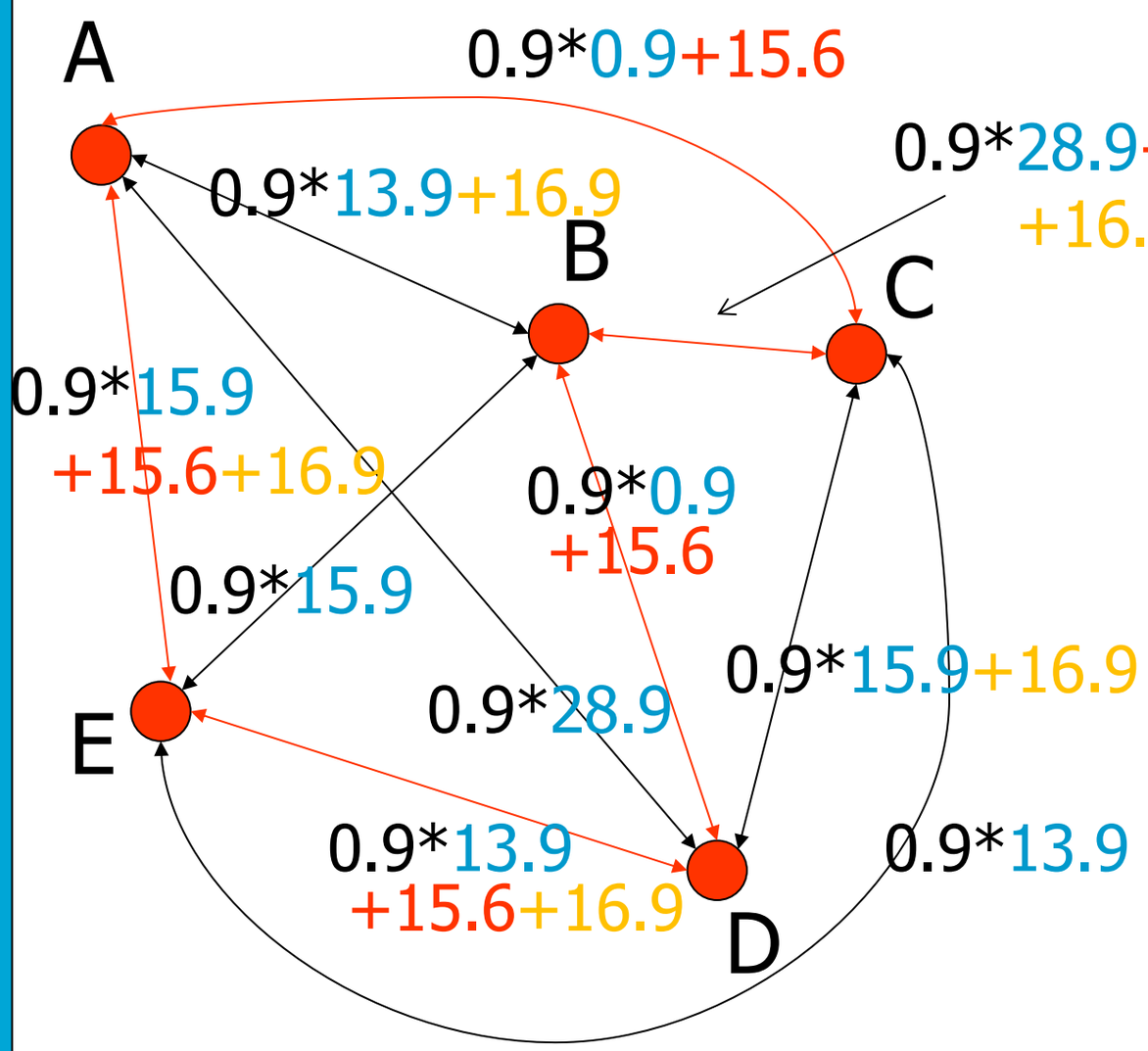
$p_{BE} = 0.944$

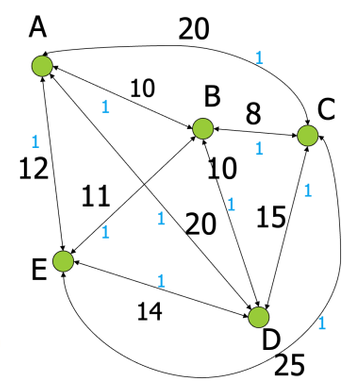
Next step would then be E and then back to A

$$p_{ij}^k(t) = \begin{cases} \dfrac{\tau_{ij}(t)^\alpha \eta_{ij}^\beta}{\sum_{m \in C_i^k} \tau_{im}(t)^\alpha \eta_{im}^\beta}, & j \in C_i^k \\ 0, & \text{otherwise} \end{cases}$$

$\alpha = 1, \beta = 0.5$

Ant Colony Optimization for TSP





Ant 1: ACBDEA	Ant 2: CBAEDC
20	8
8	10
10	12
14	14
12	15
64	59

$$\Delta\tau^k = Q \times \frac{1}{L^k}$$

$$\Delta\tau^1 = \frac{1000}{64} = 15.6 \qquad \Delta\tau^2 = \frac{1000}{59} = 16.9 \qquad \alpha = 1,\ \beta = 0.5$$

Travelling Sales Ants

- Ant after ant, shorter and shorter routes are found
- Even if a long link (which is not good for a short route) is very popular, at the end it will be neglected, because of evaporation
- Same happens for long links that are part of a short route
  - Even if in the beginning they are less likely to be selected
  - Since selection is stochastic, at some point some an ant will pass by and make the link more appealing with pheromone
- Once again, the method finds a short route, but not necessarily the shortest!

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ACO for TSP: Parameters

Parameters must be **tuned to the specific problem**

- Number of ants?
  - Typically, same as the number of cities
- Initial distribution of ants over cities
  - Uniform distribution?
- Q value
  - Large enough to not lose precision during updates
- α and β values?
- Evaporation rate?

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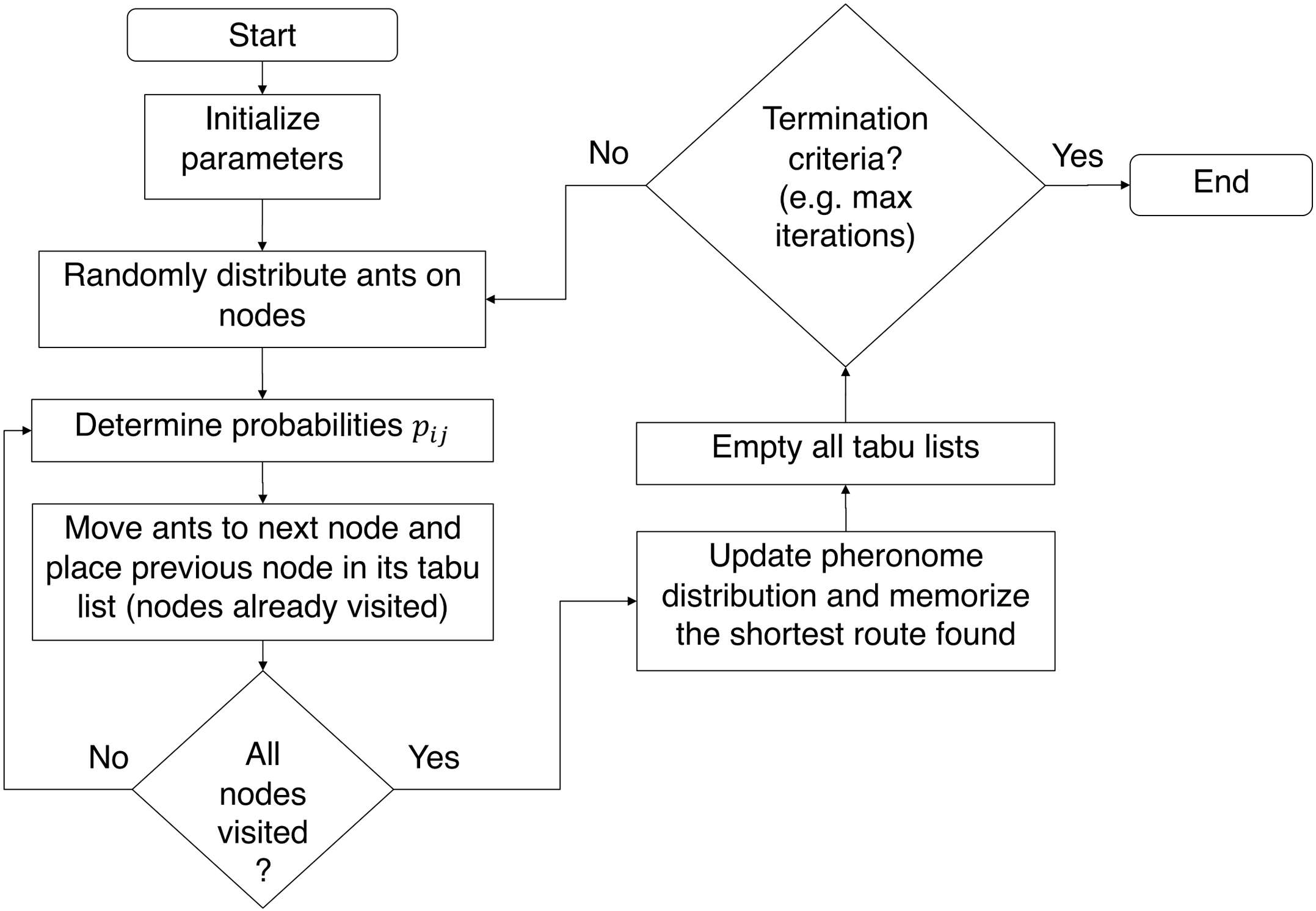
ACO Algorithm (Ant-Cycle)



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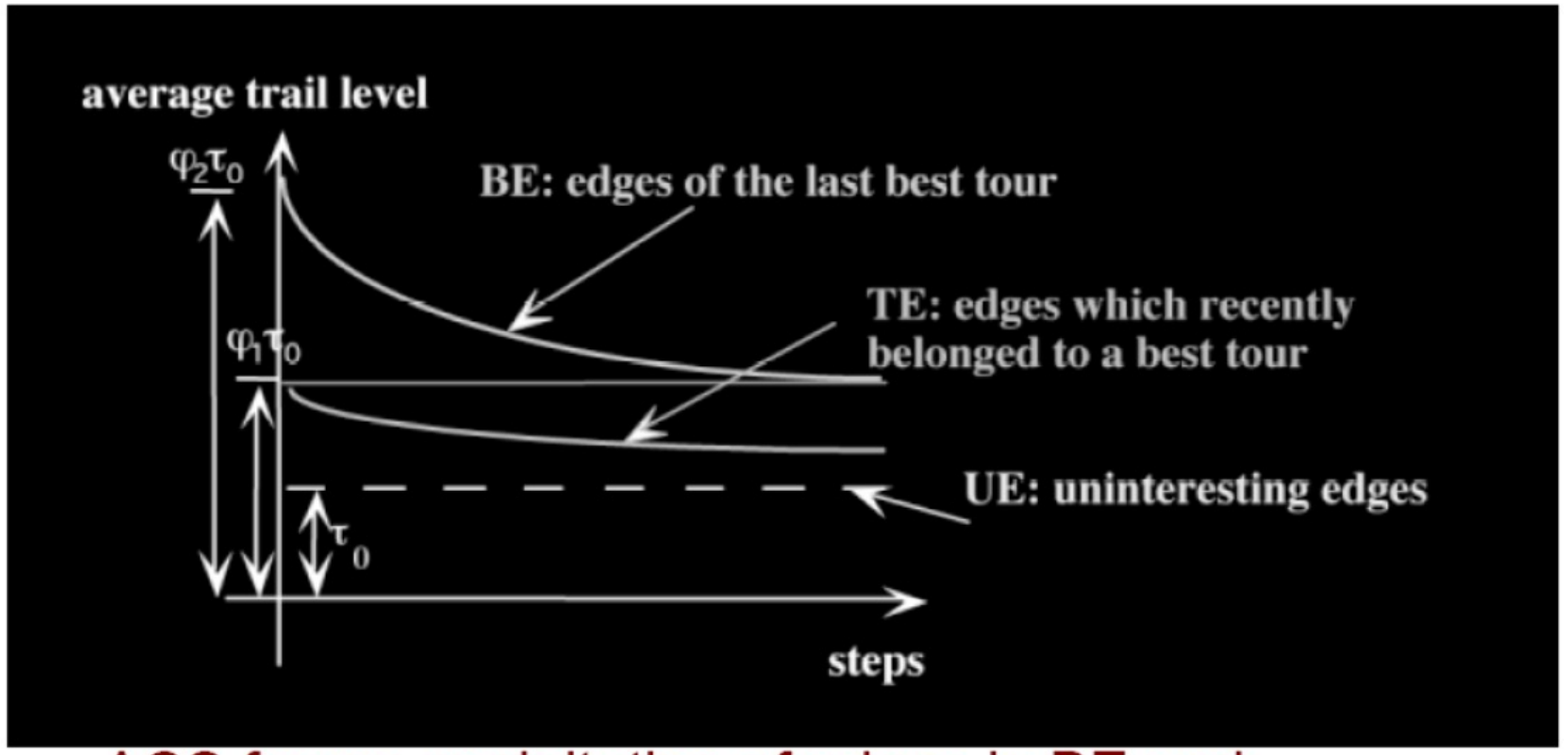
Why Does it Work?

- Three important components:
  - TIME: a shorter path receives pheromone quicker (this is often called: ”differential length effect”)
  - QUALITY: a shorter path receives more pheromone
  - COMBINATORICS: a shorter path receives pheromone more frequently because it is likely to have a lower number of decision points

https://imada.sdu.dk/u/marco/Teaching/Fall2006/DM63/Notes/dm63-lec10-slides.pdf

Analysis



ACS favors exploitation of edges in BE and exploration of edges in TE

https://imada.sdu.dk/u/marco/Teaching/Fall2006/DM63/Notes/dm63-lec10-slides.pdf

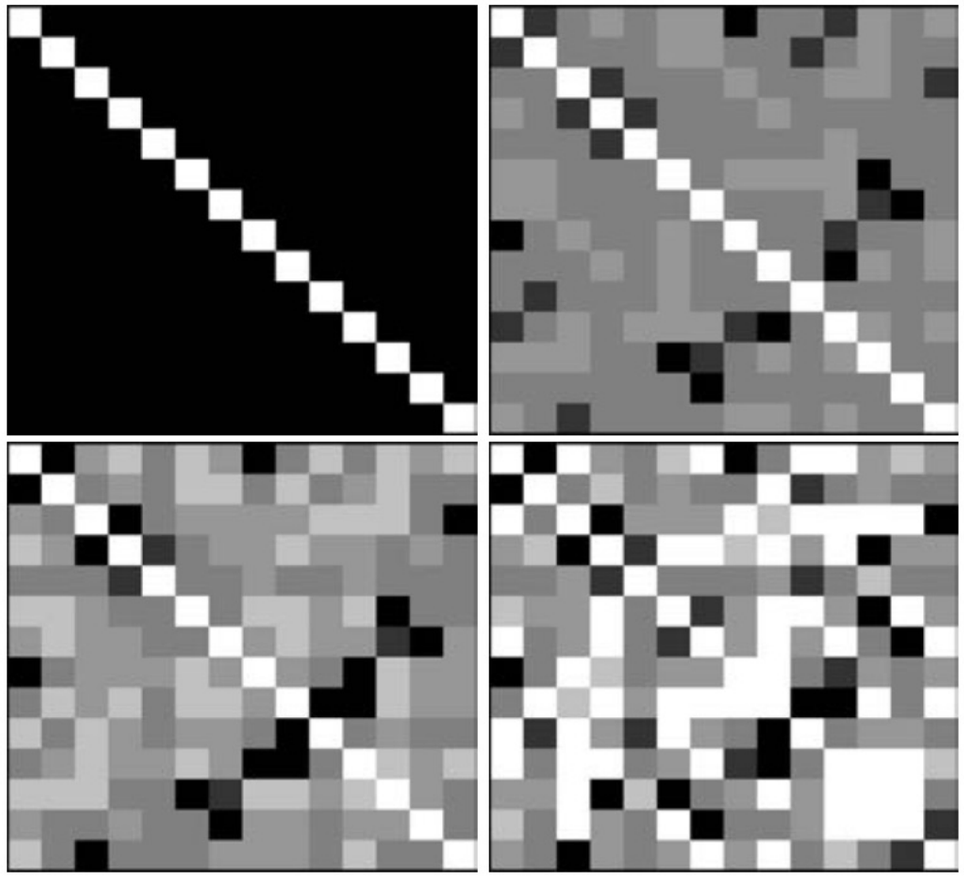


Figure 3.5
A visual representation of the pheromone matrix. The pheromone values on the arcs, stored in the pheromone matrix, are translated into gray-scale values; the darker an entry, the higher the associated pheromone trail value. The plots, from upper left to lower right, show the pheromone value for AS applied to TSPLIB instance `burma14` with 14 cities after 0, 5, 10, and 100 iterations. Note the symmetry with respect to the main diagonal, which is due to the fact that `burma14` is a symmetric TSP instance.

Dorigo, Marco, Mauro Birattari, and Thomas Stutzle. "Ant colony optimization." *IEEE computational intelligence magazine* 1.4 (2006): 28-39.

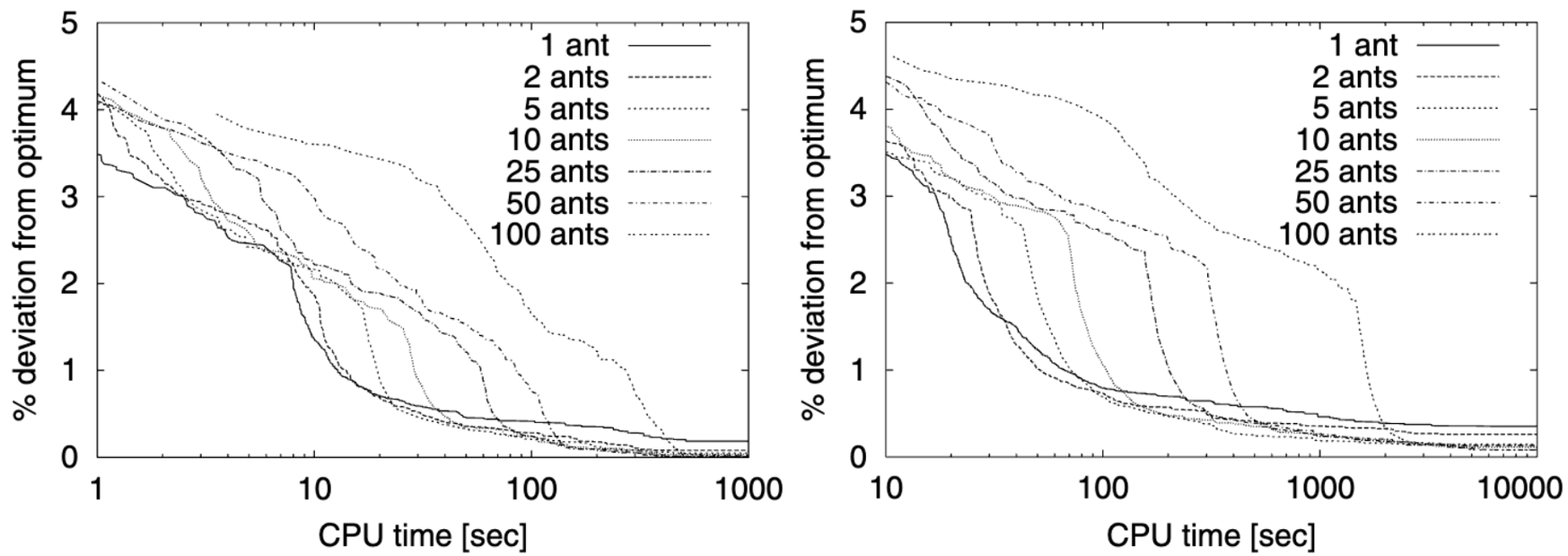


Figure 3.11
Varying the number of ants used: The plots give the average percentage deviation from the optimal tour as a function of the CPU time in seconds for $\mathcal{MM}$AS with 3-opt using a number of ants varying from 1 ant to 100 ants on the symmetric TSPLIB instances `pcb1173` (left) and `pr2392` (right). Parameters are set to the values given in box 3.3.

Dorigo, Marco, Mauro Birattari, and Thomas Stutzle. "Ant colony optimization." *IEEE computational intelligence magazine* 1.4 (2006): 28-39.

Variations on the ACO algorithm

- The Ant-Cycle algorithm achieved encouraging initial results in the TSP, but was found to be inferior to other state-of-the-art algorithms for the TSP
- Other extensions achieved better results, and also led to other applications

ACO algorithm	TSP	Main references
Ant System (AS)	yes	Dorigo (1992); Dorigo, Maniezzo, & Colorni (1991a,b, 1996)
Elitist AS	yes	Dorigo (1992); Dorigo, Maniezzo, & Colorni (1991a,b, 1996)
Ant-Q	yes	Gambardella & Dorigo (1995); Dorigo & Gambardella (1996)
Ant Colony System	yes	Dorigo & Gambardella (1997a,b)
$\mathcal{MAX}$–$\mathcal{MIN}$ AS	yes	Stützle & Hoos (1996, 2000); Stützle (1999)
Rank-based AS	yes	Bullnheimer, Hartl, & Strauss (1997, 1999c)
ANTS	no	Maniezzo (1999)
Hyper-cube AS	no	Blum, Roli, & Dorigo (2001); Blum & Dorigo (2004)

In the column TSP we indicate whether this ACO algorithm has already been applied to the traveling salesman problem.

- In these extensions, the main differences are the way the pheromone update is performed, and some details in the management of the pheromone trails

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Variations on the ACO algorithm

- Elitist Ant Systems: Dorigo (1992) and Dorigo et al., (1991a, 1996)
  - Provide additional reinforcement to the best routers found since the start of the algorithm

$$\tau_{ij} \leftarrow \tau_{ij} + \sum_{k=1}^{m} \Delta\tau_{ij}^{k} + e\Delta\tau_{ij}^{bs},$$

length

$$\Delta\tau_{ij}^{bs} = \begin{cases} 1/C^{bs}, & \text{if arc } (i,j) \text{ belongs to } T^{bs}; \\ 0, & \text{otherwise.} \end{cases}$$

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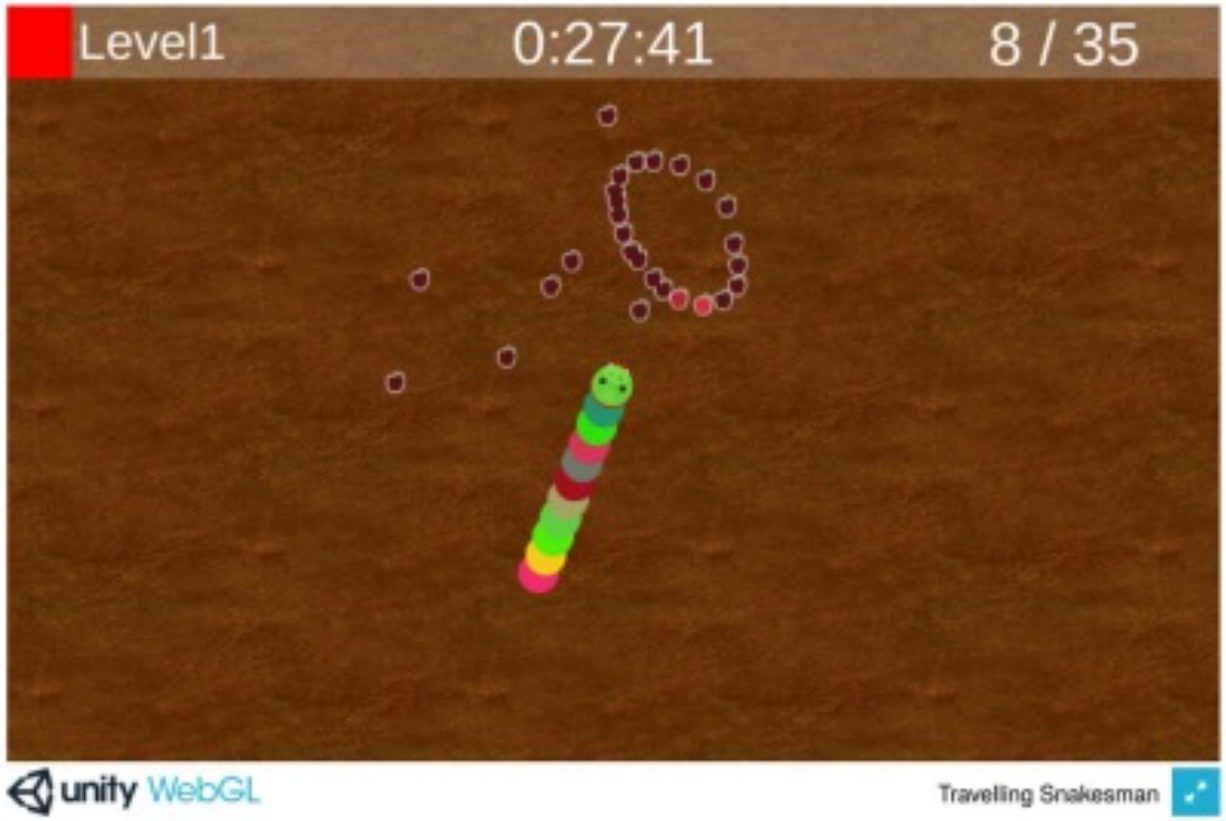
Variations on the ACO algorithm

- Extending the ACO with human intelligence
  - “Interactive machine learning: experimental evidence for the human in the algorithmic loop - A case study on Ant Colony Optimization”
  - Each time the human travels over an ant, the current pheromone value of the edge is multiplied by 5.
    - Recommendation: Link with highest probability



Holzinger, A., Plass, M., Kickmeier-Rust, M., Holzinger, K., Crișan, G. C., Pintea, C. M., & Palade, V. (2019). Interactive machine learning: experimental evidence for the human in the algorithmic loop. Applied Intelligence, 49(7), 2401-2414.
Game (Firefox): https://iml.hci-kdd.org/TravellingSnakesman/

Other applications

- Scheduling
  - Assembly lines
  - Airline crew scheduling
- Routing
  - Vehicle routing
  - Circuit boards
- Image processing
  - Edge detection and edge linking
- Network design
  - Energy, gas, internet
- Biology
  - Protein folding

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Other applications

- Airline crew scheduling

Taipei–Tokyo sample timetable – eight flights.

Flight num	Dep. city	Days of service	Dep. time	Des. city	Arr. time	Flying time
1	Taipei	Mon	10:55	Tokyo	13:50	2:55
2	Taipei	Mon	12:55	Tokyo	15:50	2:55
3	Taipei	Tue	14:55	Tokyo	17:50	2:55
4	Taipei	Wed	12:55	Tokyo	15:50	2:55
5	Tokyo	Mon	16:55	Taipei	19:50	2:55
6	Tokyo	Tue	14:55	Taipei	17:50	2:55
7	Tokyo	Wed	14:55	Taipei	17:50	2:55
8	Tokyo	Wed	18:15	Taipei	21:40	2:55

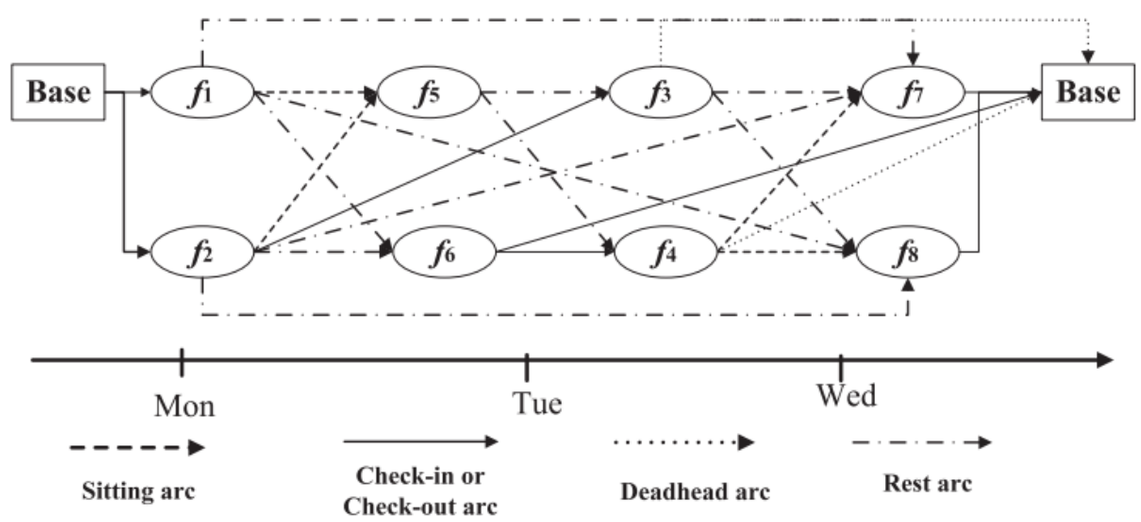


Fig. 2. Sample timetable as the search for the shortest path.

Deng, G. F., & Lin, W. T. (2011). Ant colony optimization-based algorithm for airline crew scheduling problem. *Expert Systems with Applications*, *38*(5), 5787-5793.

A bit more on two examples on other applications

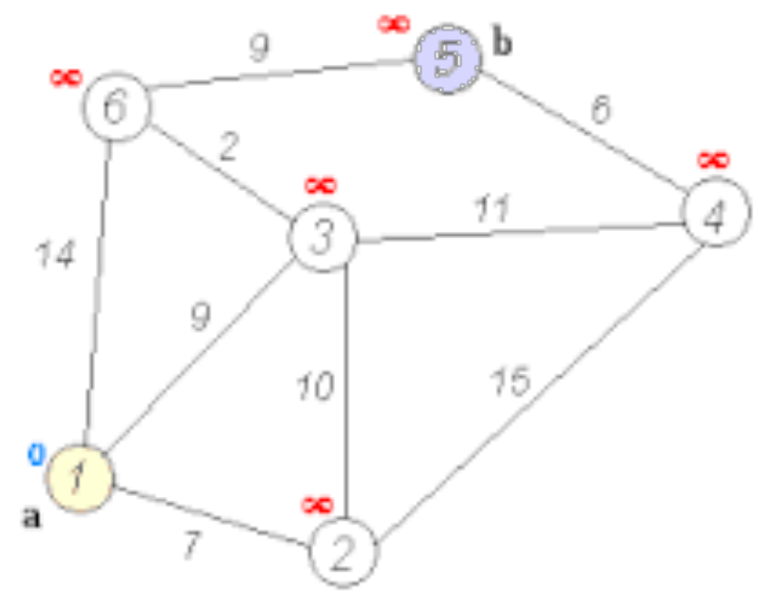
- Image processing for magnetic resonance
  - Brain segmentation to calculate brain's volume changes

Fig. 2 Results of brain segmentation. **a** Original image. **b** Divided to WM. **c** GM. **d** CSF using the proposed ACO algorithm

Khorram, B., & Yazdi, M. (2019). A new optimized thresholding method using ant colony algorithm for MR brain image segmentation. *Journal of digital imaging*, *32*(1), 162-174.

Comparison between ACO and other algorithms

- Dijkstra usually requires less computational power
- ACO is usually better suited than Dijkstra for:
  - Scenarios where you have a dynamic cost function (such as traffic in a city)
  - Very Large scenarios



Baeza, D., Ihle, C. F., & Ortiz, J. M. (2017). A comparison between ACO and Dijkstra algorithms for optimal ore concentrate pipeline routing. Journal of Cleaner Production, 144, 149-160.

Shojaie, A. A., & Seyedi Bariran, S. E. (2021). A Comparison of Extended Dijkstra and ACO Algorithm for Shortest Path Problem in Dynamic Networks with Delay Times. *Advances in Industrial Engineering*, *55*(1), 1-26.

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Ant Colony Optimization: Summary

- A big advantage of ACO is its inherent flexibility
- Environment and conditions might be unknown and/or dynamic (changing over time)
  - Internet: node breaks down
  - Highway: closed for accident
- Ant keep exploring: they adapt fast
- Pheromone gives backup options
  - If a link is closed, the second shortest route is easy to reconstruct

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